

Week 1: Tilings. Facilitator guide (W1-F-v1)

This guide is for the three adults: the organizer (math PhD), the second mathematician, and the parent volunteer. It contains everything needed to run the hour: materials, the launch, the order of problems at each table with timings, the answer and a complete argument for every problem on the three student packets, hints in the order to give them, what to do when a table stalls or finishes early, and a short section on the mathematics for the two mathematicians. Nothing in it needs to be worked out during the hour.

The student packets are **W1-K-v1** (grades K–1, 19 problems on 10 pages, the adult reads aloud and writes), **W1-M-v1** (grades 2–3, 26 problems on 15 pages) and **W1-U-v1** (grades 4–5, 30 problems on 21 pages). Each packet is deliberately longer than an hour. The *core* sequence named in each table's order line is about the first 40 minutes of work and ends with the finding that table brings to the share; everything after it is *reserve*, and most tables will not reach the reserve. Rough timings are given below; it is fine to end anywhere.

1. The session at a glance

- 0:00–0:05 Free handling. Children sit at their table and play with the pieces. K–1: build anything with pattern blocks. Grades 2–3 and 4–5: cut a few spare pieces from the cut sheets, try laying dominoes on any board.
- 0:05–0:08 Whole-group launch (Section 3), led by the organizer.
- 0:08–0:50 Table work. Core sequence first (it fills the 40 minutes for most tables), reserve problems after. Adults keep the packet moving: act, record, compare, then ask why.
- 0:50–0:58 Share (Section 8): each table shows one finding and one reason.
- 0:58–1:00 Pieces back in the bags; packets go home.

Suggested seating: the parent volunteer at K–1 (the K–1 answers and arguments below are complete and short; the organizer visits this table at the two “why” moments, Problems 6 and 11), the second mathematician at grades 2–3, the organizer at grades 4–5.

2. Materials and preparation

Printing. Print every student page at 100% (“Actual size”), not “Fit to page”: the K–1 outlines must match the 1-inch pattern blocks and the square boards must match the paper pieces. Check one page with a ruler: a K–1 hexagon is 2.0 inches corner to corner; a grades 2–3 square is 1 inch; a grades 4–5 square is 0.8 inch.

K–1 packet	3 copies plus 1 spare. Pages 2, 3, 4 (colouring pages) may need a second copy for a child who wants to redo one.
2–3 packet	4 copies plus 1 spare.
4–5 packet	3 copies plus 1 spare.
Cut sheet, 1 inch	6 copies of the fourth page from the end of this guide (one per child at grades 2–3, two for the K–1 table, whose three children need three black squares); each gives 15 dominoes, 10 L's, 2 black squares.
Cut sheets, 0.8 inch	3 copies each of sheets (a) and (b), the third and second pages from the end of this guide: per child 24 L's, 10 straight trominoes, 21 dominoes, 2 black squares, 9 straight tetrominoes, 4 L-tetrominoes. Sheet (c), the last page, is needed for Problem 28 of the 4–5 packet (nine L-tetrominoes per child; sheet (b) gives only 4): 12 L-tetrominoes and spares; print 3 copies if that problem is in reach.

Cutting. Cut the pieces before the session (cutting 24 L's takes a child ten minutes). Cut along the thick lines; the light grey lines are the squares. Keep each child's pieces in a small bag or envelope. Pieces must come from a sheet of the same scale as the board they are used on: 1-inch pieces for the 2–3 packet and K–1, 0.8-inch pieces for the 4–5 packet.

Per table.

K–1 (3 children, 1 adult). Pattern blocks, at least: 48 green triangles, 36 blue rhombi, 24 red trapezoids, 9 yellow hexagons, 6 purple chevrons (the V-shaped piece equal to two blue rhombi joined; if your set's "chevron" is a different shape, leave it out; only the purple parts of Problems 1, 2, 5, 12 and 19 mention it). Crayons in green, blue, red, yellow, purple and three other colours. From the two 1-inch cut sheets: 5 dominoes and 1 black square per child (the two sheets give 30 dominoes and 4 black squares). The adult keeps this guide open at Section 5.

Grades 2–3 (4 children, 1 adult). Per child: 15 dominoes, 10 L's, 2 black squares (one 1-inch cut sheet), a pencil and three coloured pencils.

Grades 4–5 (3 children, 1 adult). Per child: the pieces from 0.8-inch sheets (a) and (b), a pencil and three coloured pencils. For Problems 26–27: 12 blue rhombi per child from the pattern-block set.

Launch. One 2 by 3 board and one 3 by 3 board with 1-inch squares (pages 1 and 3 of the 2–3 packet, or draw them on a whiteboard), four 1-inch dominoes and one black square.

3. The whole-group launch (two to three minutes)

Everyone stands where they can see. The organizer holds up the 2 by 3 board and three dominoes and lays them on it.

"This is a tiling: the dominoes cover the board with **no gaps, no overlaps, nothing sticking out**. Every table today is playing this game with different pieces and different shapes."

Hold up the 3 by 3 board. "Can I cover this one with dominoes?" Take the answers. "Count: nine squares. Each domino covers two. Four dominoes cover eight squares, so one square is always left over. That is a *no* with a reason. 'I tried and it didn't work' is never a reason; 'it is nine squares and dominoes cover two at a time' is."

Put the black square on the centre. "A black square is a hole. A piece may **never lie on the black square**. Eight white squares, four dominoes. Can you?" Show it, or leave it for the tables.

“Three things to find out today: *Can it be done?* If yes, show us. If no, tell us why. And when there are several ways, *how many?* Two ways are different if some piece lies in a different place on the paper. Go.”

4. Rules to repeat at every table

- No gaps, no overlaps, nothing sticking out, never on the black square. Pieces lie on the grid lines and cover whole squares, never diagonally.
- Pattern blocks must meet corner to corner. A trapezoid slid half a unit leaves a sliver nothing can fill; you see it as a piece poking out of the outline.
- An L may be turned *and* flipped over (four positions in a 2 by 2 window). A straight tromino has two positions. A child who uses one orientation only will conclude “impossible” too early.
- “Different”: two fillings of the same printed board are different if some piece lies in a different place. A turned or mirrored filling counts separately. That is what 1, 2, 3, 5, 8, 11 and “two ways with three blues” count. If a child says “that’s the same one, just turned”, answer: on this fixed board it is a different filling.
- “Sets of pieces” (K–1 Problems 5 and 19) is a different count from fillings: the yellow hexagon has 10 sets (8 if the purple chevron is left out) but 52 fillings. Keep K–1 on sets.
- A 2 by 2 minus one square is covered by *one* L. Some children object that one piece is not a tiling. Say: we use one piece, and that counts.
- A failed attempt is not a proof of no. Every “no” on the pages has an argument below; do not let “we tried and couldn’t” stand.

5. K–1 table

Order and timing (core, about 55 minutes; a table that finishes Problem 12 has done the session). Problems 1–2 (7 min), 3–4 (8), 5 (12–15), 6–7 (6), 8–10 (10), 11–12 (11). Problem 5 is the long one: after six sets ask for a set with a purple and a set with no red, then move on. **Reserve:** 13–14 (15), 15–16 (8), 17–18 (12), 19 (6). If time is short, drop the second way in Problem 12 and the colouring in Problem 10.

How to run it. Read each problem aloud, one sentence at a time, while the child does it. Write the numbers in the boxes yourself; circle YES or NO yourself after the child says it. Let the three children work on their own copies at their own pace; when two disagree (“blues can’t” / “blues can”), have them show each other. The “why” questions below are for after the child has recorded what happened; ask the child to point at the picture while answering.

Problem 1.

Answer. Blue: 2 greens. Red: 3. Purple: 4. Yellow: 6.

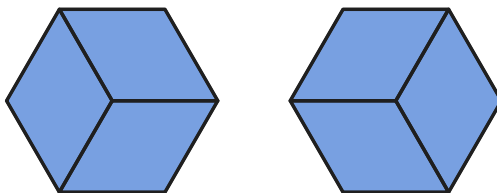
Notes. Every later count is “how many greens”. If a child places greens with gaps, run a finger along the outline: nothing sticks out, no white shows.

Problem 2.

Answer. Yellow with blues: 3. Yellow with reds: 2. Purple with blues: 2.

Problem 3.

Answer. Exactly **2 ways** with three blues. The second is the first turned by one sixth of a turn; side by side they look like a cube seen from two directions.

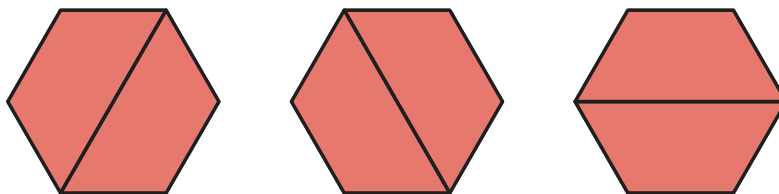


Argument. Every blue inside the hexagon uses two neighbouring triangles around the centre. The first blue can sit in 6 places; once it is placed, the remaining four triangles form a V, which two blues fill in exactly one way. Three blues make a filling, and each filling arises from three of the six starting positions, so there are $6/3 = 2$ fillings. A child's version: "Put the first blue anywhere; the rest is forced. Two pictures come out."

Hints, in order. (1) "Is there a blue standing where you had one lying?" (2) Turn the child's finished hexagon by one click and ask whether the blues are in the same places. (3) Ask them to try a third way and to say what always happens.

Problem 4.

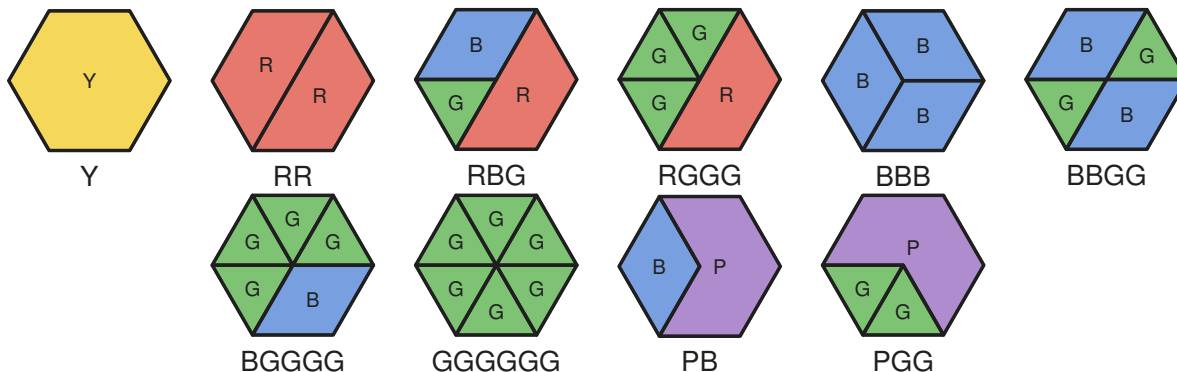
Answer. **3 ways** with two reds: the line between the two reds is one of the three long diagonals of the hexagon.



Argument. A red inside the hexagon is three neighbouring triangles around the centre, so it is one half of the hexagon; the other half is the other red. The dividing line is a long diagonal, and there are three of them.

Problem 5.

Answer. 10 sets: Y; RR; RBG; RGGG; BBB; BBGG; BGGGG; GGGGGG; PB; PGG. Without the chevron, 8 (drop PB and PGG). (The number of *fillings* is 52, or 40 without the chevron; keep the children on sets.) One filling of each set:



Argument. Sort by the biggest piece. Yellow: one set. Biggest is a red: a red is half the hexagon; the

other half is a red, or a blue and a green, or three greens: three sets. Biggest is a blue: the rest is a V of four triangles: two blues, or a blue and two greens, or four greens: three sets. Only greens: one set. $1 + 3 + 3 + 1 = 8$. If the purple is in play: biggest is a purple, the rest is a rhombus: a blue or two greens, two more sets.

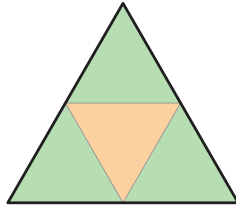
Hints, in order. (1) “What is the biggest piece you used? Can you find a filling with a red in it? With no red?” (2) “You have a red and a blue and a green. Is there a filling with a red and no blue?” (3) “Have you found every set with a red in it? How do you know?” (this is the case argument above; let them say it as “the other half is a red, or a blue and a green, or three greens”).

Notes. The child should colour the triangles with the colour of the block: the record is which pieces were used, not where.

Problem 6.

Answer. Greens: 4. Blues only: **NO**.

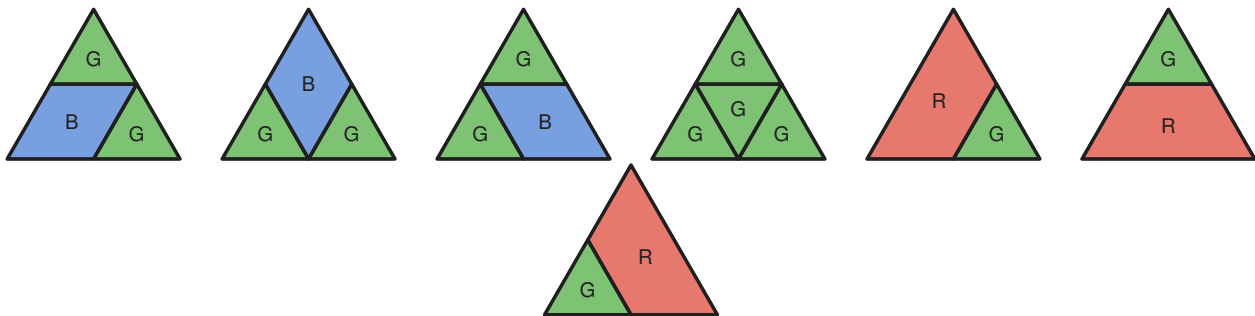
Argument. The four little triangles are not alike: three point up like mountains and one points down. Colour a copy that way (mountains green, the valley orange) or say it aloud while pointing. Put a blue anywhere on the triangle: it always covers one mountain and one valley. Two blues would need two valleys; there is one. This is the K–1 checkerboard argument; it is used again in Problems 8, 11 and 12.



Hints, in order. (1) “Put one blue down. What is left? Can a blue sit on it?” (2) “What does a blue always cover: point to the two triangles under it.” (3) “How many point up? How many point down?”

Problem 7.

Answer. 3 ways: the green sits on one of the three corners and the red fills the rest.



(The picture also shows the other fillings of this triangle, one blue and two greens in 3 ways and four greens, for a child who asks.)

Argument. The green is on one of the four little triangles. If it is the middle (down-pointing) one, the three mountains left are not a red (they do not touch along whole sides), so the green is on a corner: three corners, three ways, and each leaves a trapezoid.

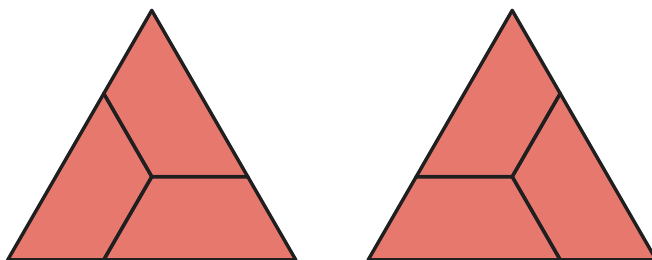
Problem 8.

Answer. Greens: 9. Blues only: **NO**. Reds only: **YES** (the pinwheel, Problem 9).

Argument. Blues: nine greens cannot be paired up; “nine is one too many for pairs”. Also, 6 point up and 3 point down, and each blue covers one of each. Reds: show the pinwheel.

Problem 9.

Answer. 2 ways: the pinwheel and its mirror image.

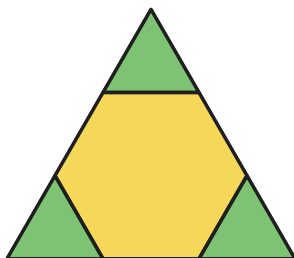


Argument. Each corner of the big triangle is a little triangle with only one neighbour, the valley next to it. The red that covers a corner must also cover that valley and then one more cell in line with them, so it runs along one of the two sides that meet at the corner. Two reds cannot run along the same side: a side holds five little triangles and a red takes three of them starting at one end, so two reds along one side would share the middle cell. Three corners, three sides, one red per side, and each red starts at one of the two corners of its side: either every red starts at the corner clockwise from its side or every red starts at the counterclockwise corner. That is the pinwheel and its mirror, and both do cover the whole triangle.

Hints, in order. (1) “Put one red along the bottom. Where can the second one go?” (2) “Is your second picture really different, or the same one turned?” (turn the child’s paper: the mirror cannot be turned into the original).

Problem 10.

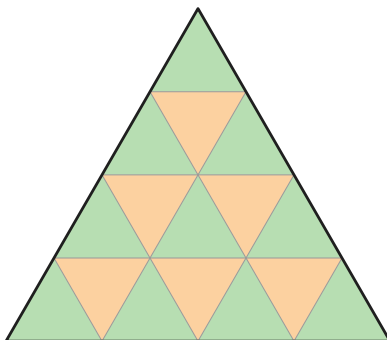
Answer. The yellow fits in one place only, around the centre point (the only corner point inside the side-3 triangle; a yellow always sits around a point where six triangles meet). Two yellows: **NO** (there is only one place, and two would need 12 little triangles of the 9). Greens: 3, at the three corners; one way.



Problem 11.

Answer. Blues only: **NO**. Reds only: **NO**.

Argument. Blues: the triangle has 16 little triangles, an even number, and still blues cannot do it: 10 point up and 6 point down. Every blue covers one of each, so after 6 blues at most, 4 mountains are always left, and no blue can lie on them. Reds: 16 is not a multiple of 3 ($5 \times 3 = 15$, $6 \times 3 = 18$).

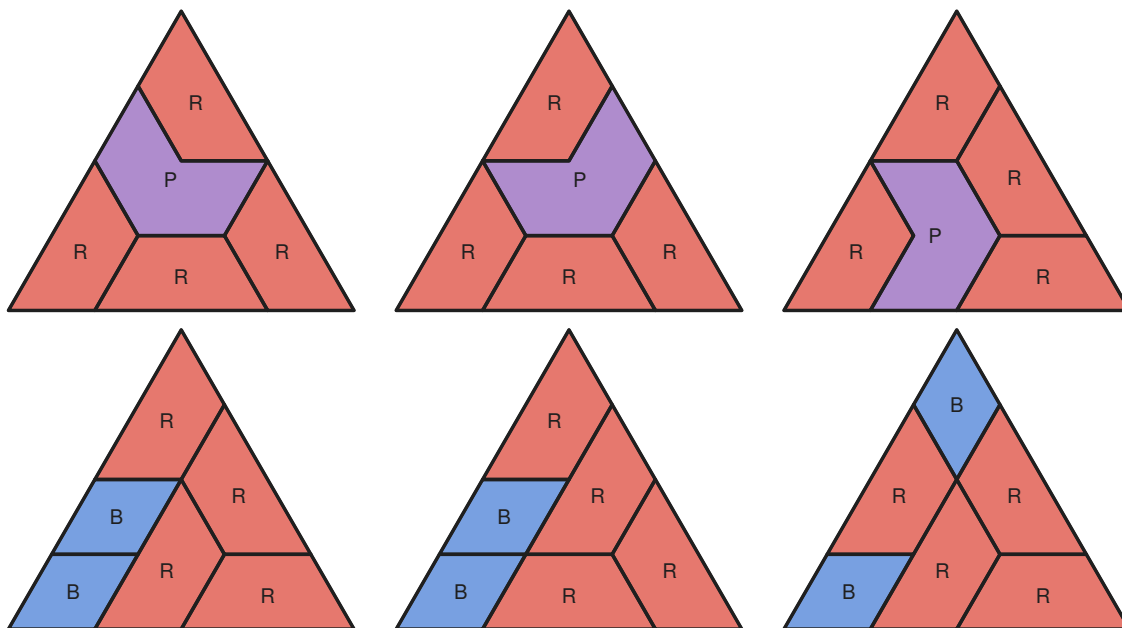


Hints, in order. (1) “Put down as many blues as you can. What is left? What do the leftover triangles all have in common?” (2) Colour the mountains on the small copy and count both kinds.

Notes. This is the moment the organizer should visit: “even, and still impossible” is the point of the whole session (the same surprise as the checkerboard with two corners removed).

Problem 12.

Answer. The only ways with no green: **4 reds and 1 purple** (6 fillings) or **4 reds and 2 blues** (39 fillings). The number of reds is always 4.

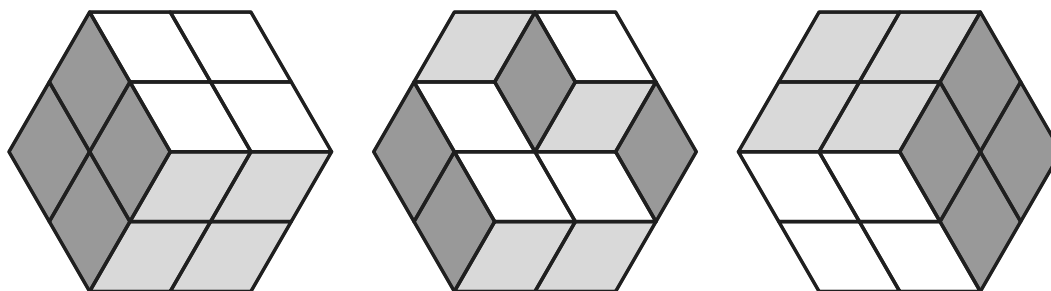


Argument. Blue, purple and yellow each cover as many mountains as valleys (1 and 1, 2 and 2, 3 and 3). Only a red is lopsided: 2 mountains and 1 valley, or 1 and 2. The big triangle has 4 more mountains than valleys, so it takes at least 4 reds, and 4 reds only if all four lie long side down. Four reds cover 12 little triangles; the 4 left over are two mountains and two valleys, filled by a purple or by two blues. More than 4 reds is impossible: 5 reds cannot make a difference of 4 (five odd numbers add to an odd number), and 6 reds are 18 triangles, too many.

Hints, in order. (1) “Which piece did you use most? Try with only three reds.” (2) “Put the four reds long side down first. What shape is left?”

Problem 13. (reserve)

Answer. 12 blues. Whatever the filling, **4 standing, 4 leaning left, 4 leaning right**. There are 20 fillings in all; three of them, with the standing rhombi shaded dark and the leaning-right ones light:



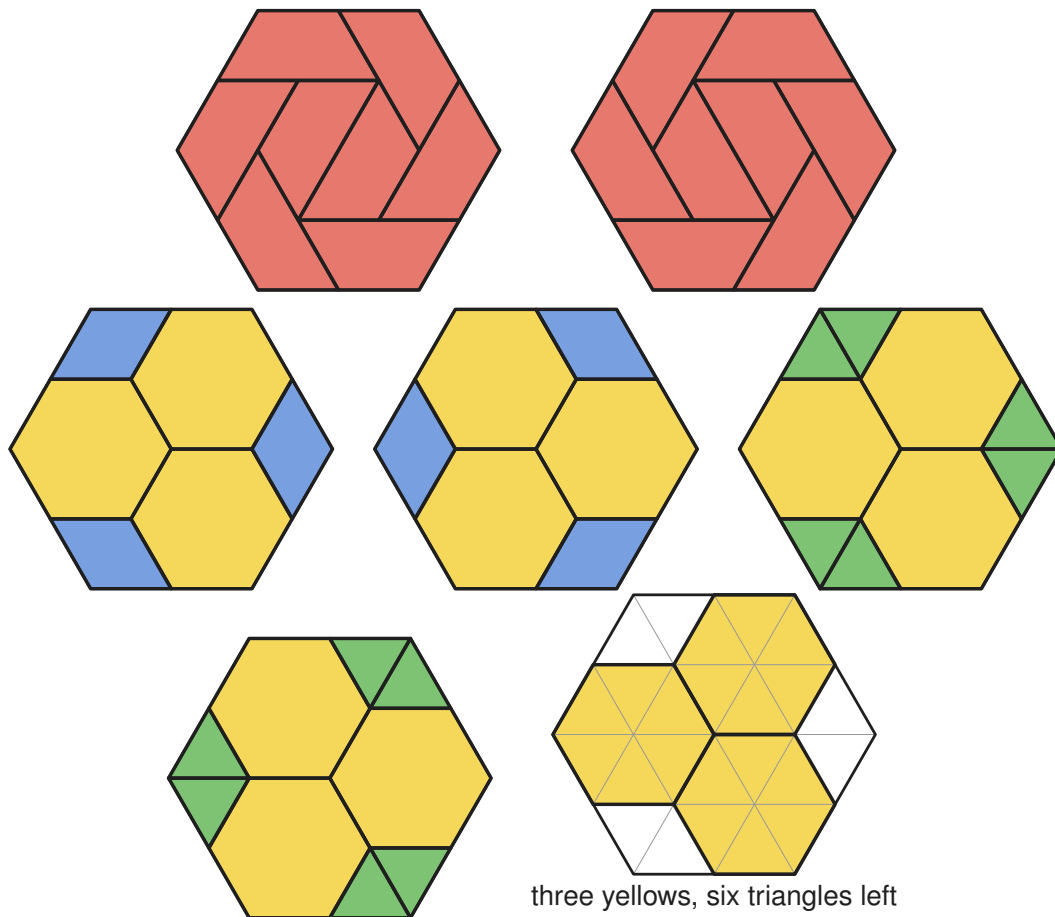
Argument. Keep the hexagon flat side down. A leaning blue has a flat bottom edge and a flat top edge, one row up; a standing blue has no flat edge. Start on one of the two little edges along the bottom of the

hexagon: the blue on it must be leaning (a standing blue has no flat edge to put there). Its flat top is a new floor, one row up, and the blue on that floor is leaning too. After four rows you are at the top. So each bottom edge starts a chain of four leaning blues, chains cannot share a blue, and every leaning blue is in a chain (follow its floor downward). Two chains of four: 8 leaning blues, so $12 - 8 = 4$ standing. Turn the picture one third of a turn and the same story counts each leaning kind: 4 and 4.

Hints, in order. For K-1 the finding (always 4, 4, 4) is the point; the chain argument is for the share if the organizer is there. (1) "Find a blue touching the bottom edge. What kind is it?" (2) "Follow it upward: what is on top of it?"

Problem 14. (reserve)

Answer. 8 reds (there are 9 fillings; two are shown). Yellows: at most **3** fit, so all-yellow is **NO**. With three yellows the six leftover triangles are three blues (2 ways) or six greens (2 ways).



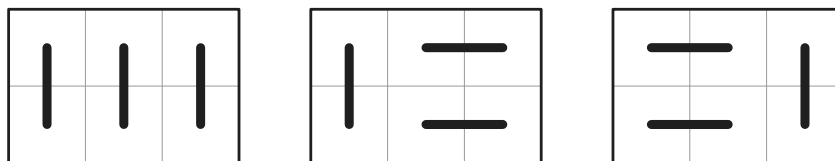
Argument. A yellow sits around a corner point inside the big hexagon; there are 7 such points, the centre and six around it in a ring. Two yellows around neighbouring points overlap (they share two triangles). The centre point neighbours all six ring points, so either one yellow at the centre and none else, or yellows on the ring with no two neighbours: at most 3, every other point. Three yellows cover 18 of the 24 triangles; 4 yellows would need 24 and never fit.

Hints, in order. (1) For a red-only filling: "Lay a red along one edge of the hexagon, long side on the edge, and go round"; the six edge reds leave the small hexagon in the middle, which two reds fill. (2) For three yellows: "Leave the middle empty."

Problem 15. (reserve)

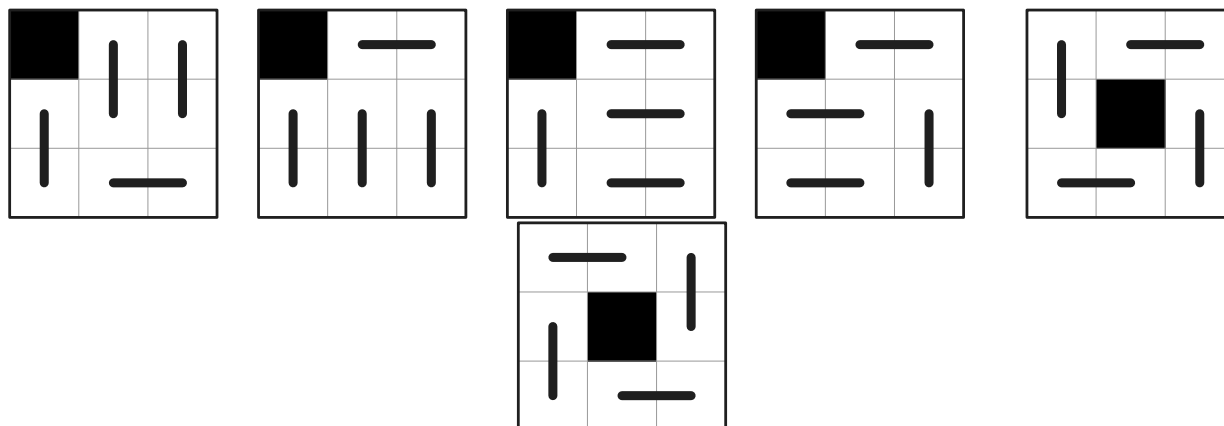
Answer. 3 ways. Why exactly 3: the left column is a standing domino (then the 2 by 2 that is left has 2

ways) or two lying dominoes, one above the other (then the last column is one standing domino: 1 way).

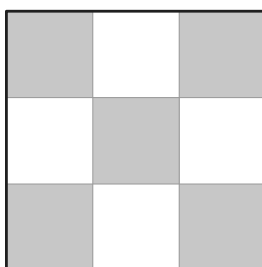


Problem 16. (reserve)

Answer. Whole board: **NO** (nine squares, dominoes cover two at a time). Corner black: **YES** (4 ways). Middle of a side black: **NO**. Centre black: **YES** (2 ways).



Argument. For the side square: colour the board like a checkerboard with the corners dark: 5 dark squares and 4 light. A domino always covers one dark and one light. Four dominoes cover four of each. If the black square is light, 5 dark squares and 3 light are left, and the dominoes cannot do it. If the black square is a corner or the centre (dark), 4 and 4 are left, and it works. This is the same reason as the mountains and valleys in Problem 6, in the square world; use it at the share.



Problem 17. (reserve)

Answer. Two greens: 1 shape (the blue rhombus). Three greens: 1 shape (the red trapezoid). Four greens: **3 shapes**: the big triangle, the straight strip (two blues in a row), and the chevron (the purple).



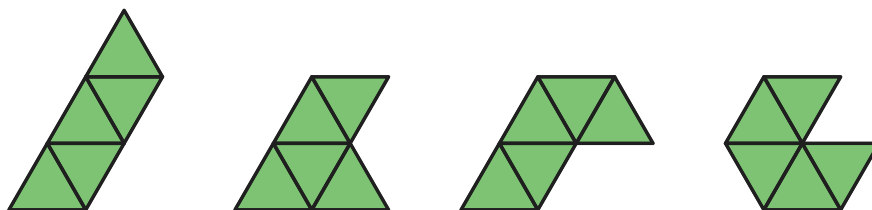
Argument. Three greens make only the red, since the third green must touch the first two along a side. Add a fourth green to a red: the red has five sides. Under the long side, either end: the chevron. On the short top side: the big triangle. On a slanted side: the strip. Three shapes, and nothing else, because every shape of four contains a shape of three, which is a red.

Hints, in order. (1) “Make the red first. Where can the fourth green go?” (2) “Is that new, or the strip

turned over?”

Problem 18. (reserve)

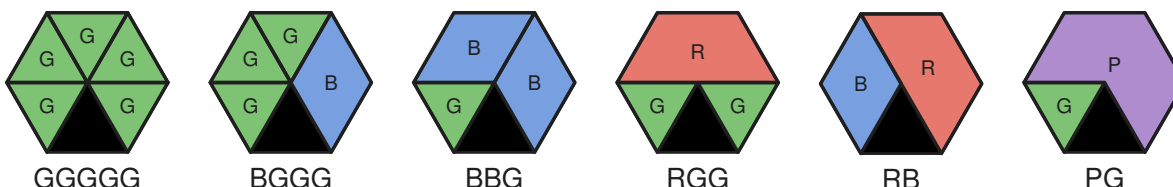
Answer. Five greens: **4 shapes**.



Argument. Every shape of five contains a shape of four; add a green to each of the three shapes of four in every place and throw away repeats. Let the child do this with blocks; the adult keeps the four drawings and checks each new one against them.

Problem 19. (reserve)

Answer. 6 sets: PG, RB, RGG, BBG, BGGG, GGGGG. Blues only: **NO** (5 triangles, odd; also 2 up and 3 down). Reds only: **NO** (5 is not a multiple of 3).



If the K–1 table stalls. On Problem 5, hand the child one filling to copy and ask for a different set. On Problems 9 and 12, hand over the first piece placed correctly. If a child is done with a problem and the others are not, they may colour the outline they built on. **If it finishes early**, go to Problems 13, 15, 17 in that order; Problem 13 is the best share item for this table after the pinwheel.

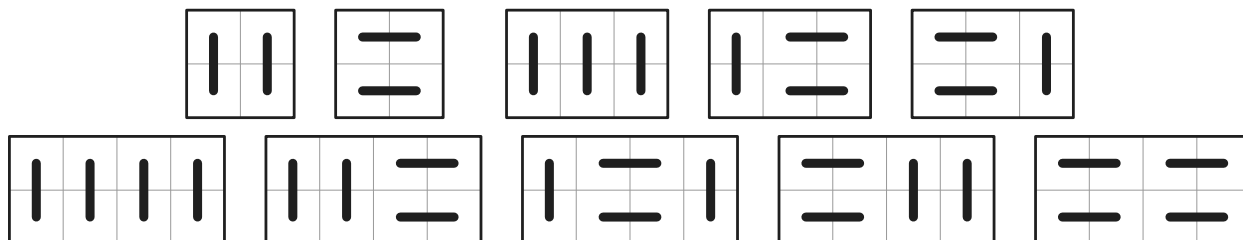
6. Grades 2–3 table

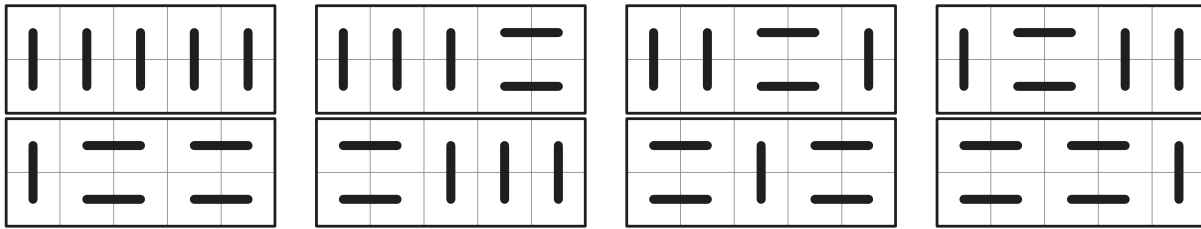
Order and timing (core, about 45 minutes). Problems 1–4 (15 min), 5–6 (6), 7–9 (10), 10 with boards A, B, C, D only (7), 12 (4; the colours of all eight boards, Y or N only for the boards done). **Reserve, in this order:** 13 (8), 14–18 (20), 10 boards E–H and 11 (10), 19–23 (25), 24–26 (18). Boards A to D of Problem 10 are enough for Problem 12; a table that reaches Problem 12 has done the session.

How to run it. The children build on the 1-inch board and record on the small boards (a thick line through the two squares of each domino; colour each L). Insist on “every way” being organised: after Problem 3 ask “how do you know there isn’t a sixth?”.

Problem 1–4.

Answer. 2 by 2: **2**. 2 by 3: **3**. 2 by 4: **5**. 2 by 5: **8**. (2 by 6: 13, 2 by 7: 21.)





Problem 5.

Answer. 1, 2, 3, 5, 8, then **13, 21**: each number is the sum of the two before it.

Problem 6.

Answer. 2 by 5: 5 boards with a standing domino at the left end, 3 without; on those the left end is two lying dominoes, one above the other; $5 + 3 = 8$. 2 by 4: $3 + 2 = 5$.

Argument. Look at the left end of a 2 by 5 tiling. Either a standing domino covers the whole left column, or a lying domino covers the top-left square; then the square under it can only be covered by another lying domino. Nothing else is possible. After a standing domino a 2 by 4 board is left (5 ways); after two lying dominoes a 2 by 3 is left (3 ways). $5 + 3 = 8$, and nothing is counted twice because the left end tells you which kind you have. The same reason gives 13 for 2 by 6 ($8 + 5$) and 21 for 2 by 7 ($13 + 8$); the table's prediction is checked without building.

Hints, in order. (1) "Look at the top-left square. What can cover it?" (2) "If it is a lying domino, what covers the square under it?" (3) "After you decide the left end, what board is left?"

Problem 7.

Answer. No: nine squares, each domino covers two, so four dominoes cover eight and one square is always left.

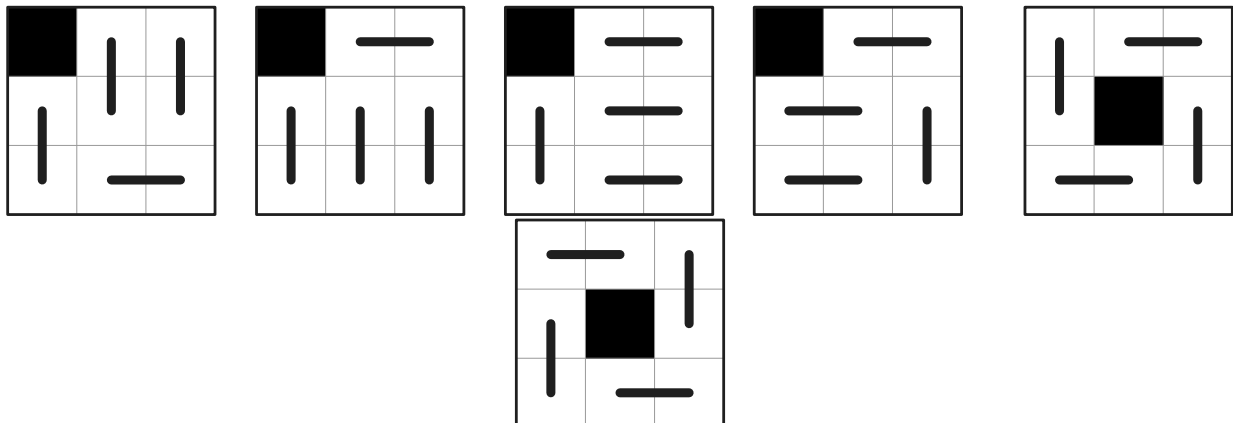
Problem 8.

Answer. The map: Y at the four corners and the centre, N at the four middle-of-side squares.

Problem 9.

Answer. Under a domino: always 1 dark and 1 light. The board has 5 dark squares and 4 light. The Y squares are dark; the N squares are light.

Argument. Four dominoes cover four dark and four light squares. If the black square is light, 5 dark and 3 light squares are left, so it cannot be done. If the black square is dark, 4 and 4 are left, and the pictures below show that it can.

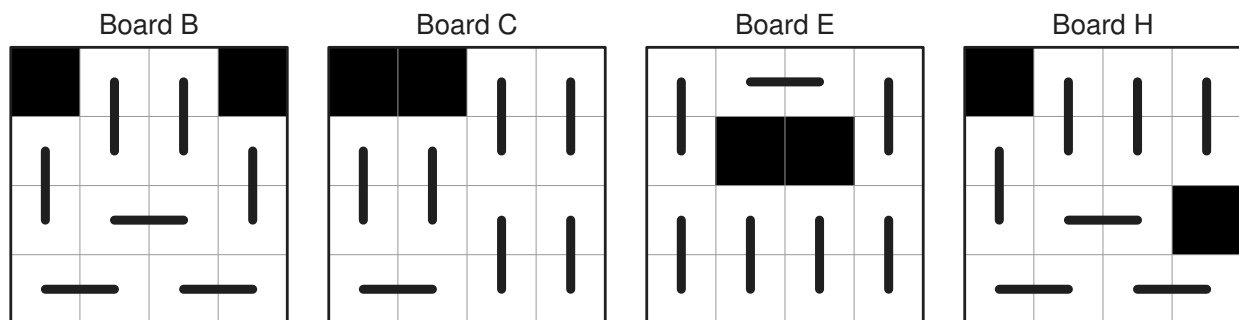


Hints, in order. (1) “Put a domino anywhere. What is under it? Move it. Always?” (2) “How many dark squares must four dominoes cover?” (3) “What colour was the black square on the boards that failed?”

Notes. Eight white squares, four dominoes, and still impossible: this is the first “no” that is not just counting squares, and children remember it. Say “the domino eats one dark and one light, so the numbers have to match”.

Problem 10. (boards A–D core, E–H reserve)

Answer. A: **N**. B: **Y** (12 tilings). C: **Y** (18). D: **N**. E: **Y** (6). F: **N**. G: **N**. H: **Y** (6). One tiling of each Y board:

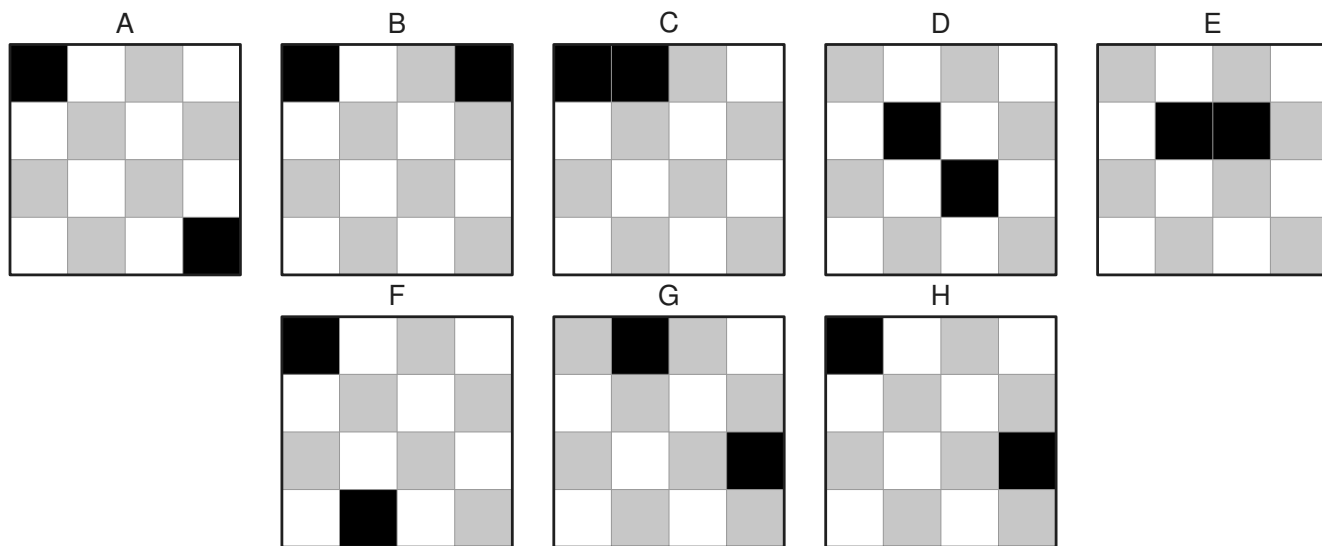


Problem 11. (reserve)

Answer. The rule: it works exactly when the two black squares have different colours on the checkerboard (one dark, one light). All 120 pairs on the 4 by 4 were checked: the 64 different-colour pairs all work, the 56 same-colour pairs never do.

Problem 12.

Answer. Colours: A dark–dark, B dark–light, C dark–light, D dark–dark, E dark–light, F dark–dark, G light–light, H dark–light.



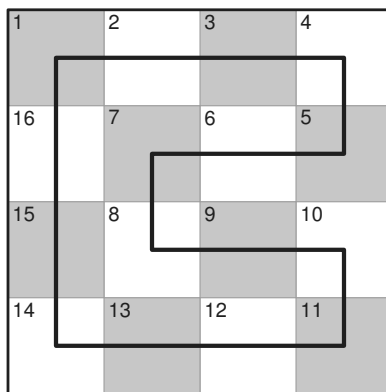
The N boards: both black squares have the same colour. The Y boards: one dark, one light.

Argument. No, when the colours are equal: 8 dark and 8 light squares, take away two dark, 6 dark and 8 light are left; 7 dominoes need 7 of each. Yes, when they differ: this is Problem 13.

Problem 13. (reserve)

Answer. With one dark and one light black square, the two roads always have an even number of squares each (for example 6 and 8, or 0 and 14 when the black squares are neighbours on the path), and

dominoes laid along a road two squares at a time cover it. Loop order for the adult, square 16 next to square 1 (this is the same loop as on the student page):

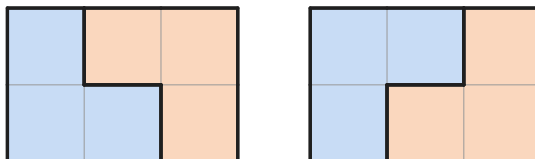


Argument. Along the path the colours alternate dark, light, dark, light. Walking from the dark black square to the light black square, the squares strictly between them start with a light one and end with a dark one, so there is an even number of them; the same for the other road. A road with an even number of squares is covered by dominoes laid one after the other along it. So every pair of one dark and one light square works, on this board and on any board with such a path (6 by 6, 8 by 8, and every even board).

Hints, in order. (1) “Follow the path with your finger from one black square to the other; count.” (2) “Lay one domino on the first two squares of the road; then the next two.”

Problem 14. (reserve)

Answer. 2 ways. Why exactly 2: the top-left square lies in some L, and that L fits in the left 2 by 2 block; of its three positions there, the two that leave a shape an L can cover are the ones drawn (the third leaves the bottom-left square cut off).

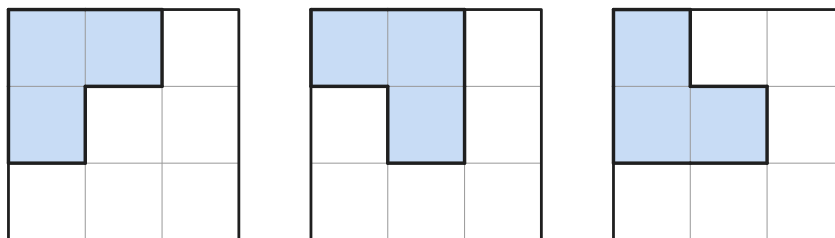


Problem 15. (reserve)

Answer. No.

Problem 16. (reserve)

Answer. Three positions cover the top-left corner (all inside the top-left 2 by 2 block). An L never covers two corners. The board has 4 corners; three L's cover at most 3 of them; a corner is always left uncovered, so Problem 15 is impossible.



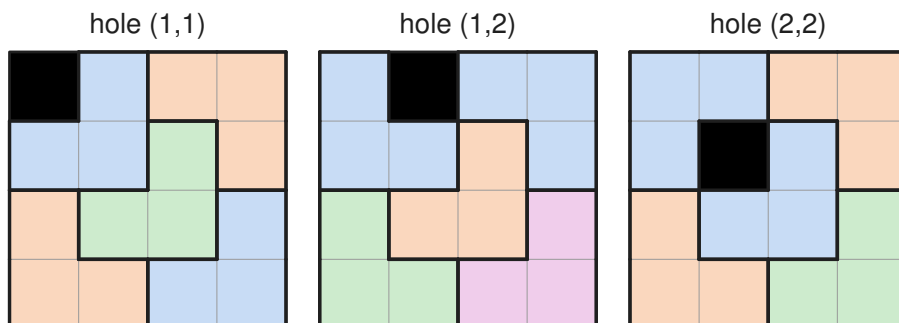
Argument. An L always fits inside some 2 by 2 block, and each 2 by 2 block of the 3 by 3 board contains exactly one corner square. So each L covers at most one corner. Four corners, three L's.

Notes. Nine squares is a multiple of three and it still fails. The dots in Problems 22–23 are the same

argument in a form that scales to 5 by 5.

Problem 17. (reserve)

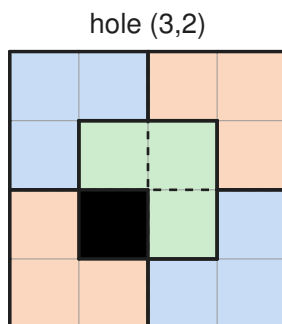
Answer. Every square works as the black square, and for every black square the tiling is the only one (checked by computer for all 16 squares; do not try to prove uniqueness at the table). Because each board has exactly one tiling, trial and error is genuinely slow here: give the middle-L hint after three minutes. P, Q, R:



Hints, in order. (1) “Put one L in the middle of the board first.” (2) For P: “the L in the middle should not touch the quarter with the black square.”

Problem 18. (reserve)

Answer. Picture T (black square at row 3, column 2) by the recipe:

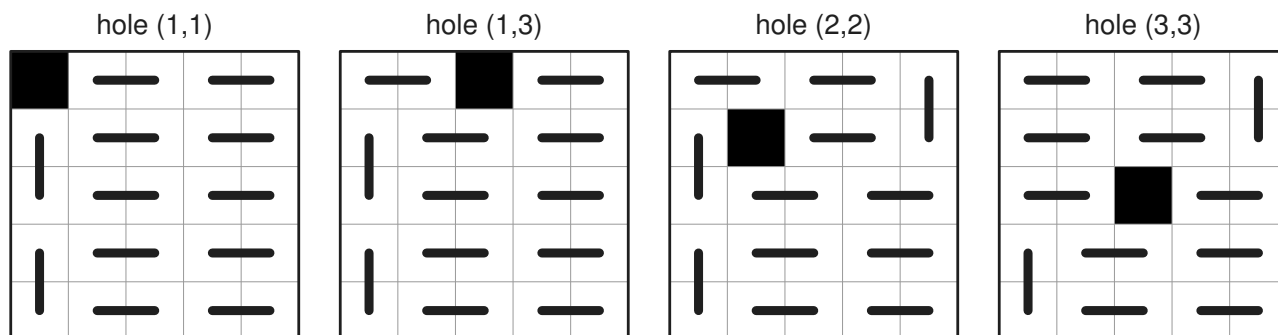


Argument. Fold the board into four 2 by 2 quarters. One quarter holds the black square. Put one L in the very middle, covering one square of each of the other three quarters. Now every quarter is a 2 by 2 with one square gone, and that is exactly one L. The recipe never needed to know where the black square is, only which quarter it is in. Three sentences: this is the argument the older table pushes to 8 by 8 and beyond.

Hints, in order. (1) “Which quarter has the black square? The middle L must stay out of it.”

Problem 19. (reserve)

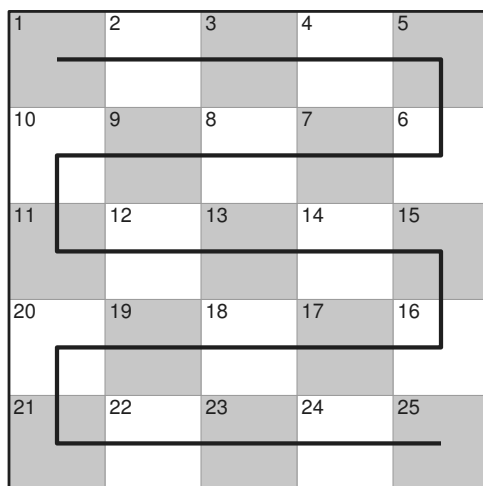
Answer. Places are given as (row, column), counted from the top-left corner. Places 1 (1,1) Y, 2 (1,2) N, 3 (1,3) Y, 4 (2,2) Y, 5 (2,3) N, 6 (3,3) Y. The full map: Y on the 13 dark squares (corners, centre, and every square with row and column both odd or both even), N on the 12 light squares. Tilings for places 1, 3, 4, 6 along the snake path:



Notes. From the 3 by 3 children guess “corners and centre only”; place 3, the middle of the top side, works and kills that guess; place 2, next to it, fails.

Problem 20. (reserve)

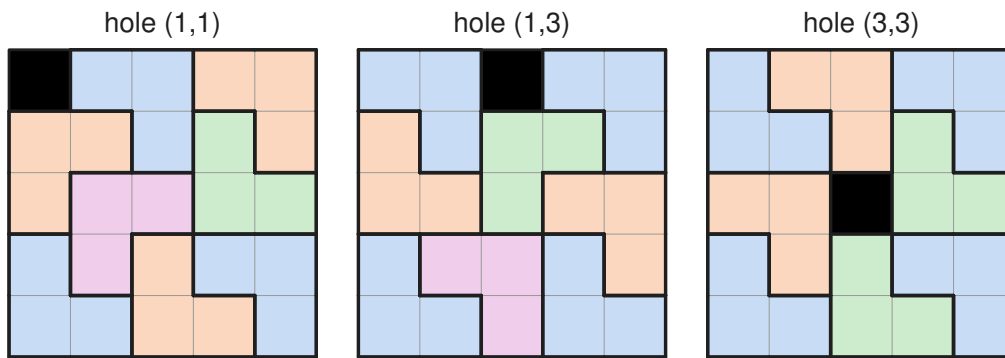
Answer. 13 dark, 12 light; the Y squares are the dark ones. Black on a dark square: both roads even (0 and 24 for the corner, 8 and 16 for square (2,2)). Black on a light square: both roads odd, and the last square of each road cannot be covered.



Argument. Twelve dominoes cover 12 dark and 12 light squares, so the black square must be dark. Along the snake the dark squares sit at the odd positions 1, 3, 5, . . . , 25; removing an odd position leaves two runs of even length (before it: an even number, after it: 25 minus an odd number, even). Dominoes laid along each run cover it. So dark squares always work and light never do; the same reason works on 7 by 7 (25 dark work, 24 light fail).

Problem 21. (reserve)

Answer. Places 1 Y, 2 N, 3 Y, 4 (2,2) **N**, 5 N, 6 Y. The L-map: Y exactly on the 9 squares with odd row and odd column, N everywhere else. Compared with the domino map: the 9 dot squares are Y on both; the four squares (2,2), (2,4), (4,2), (4,4) are Y for dominoes but N for L's. Tilings for places 1, 3, 6 (8, 16 and 32 tilings exist):

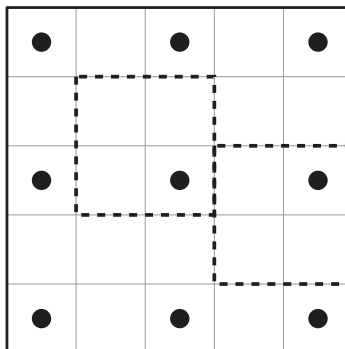


Problem 22. (reserve)

Answer. An L covers at most 1 dot; three L's at most 3; there are 4 dots, so Problem 15 is impossible (same as Problem 16, now in a form that grows).

Problem 23. (reserve)

Answer. Every 2 by 2 square contains exactly 1 dot. An L covers at most 1 dot. Eight L's cover at most 8 dots; the board has 9; so the black square must be on a dot. That is the L-map of Problem 21 exactly.

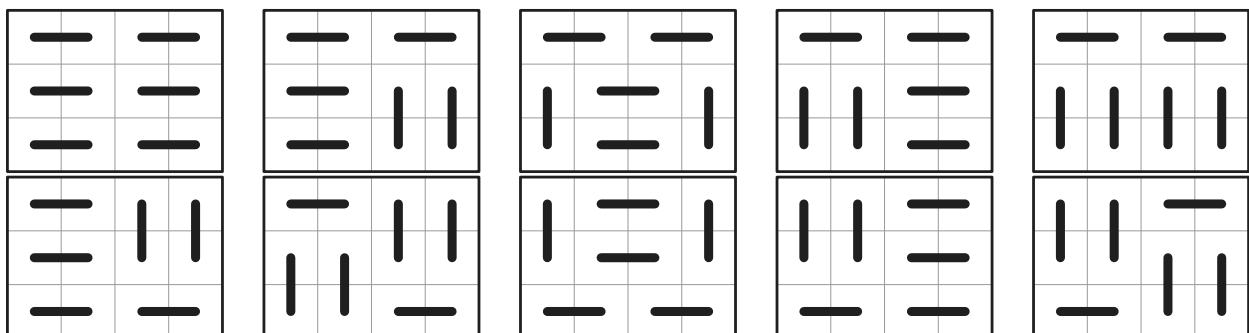


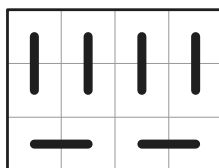
Argument. Of two neighbouring rows one is odd, and of two neighbouring columns one is odd, so a 2 by 2 window holds exactly one (odd, odd) square. Every L fits in a 2 by 2 window, so it covers at most one dot. If the black square were not a dot, 8 L's would have to cover 9 dots.

Notes. The dots are not the checkerboard (13 dark squares); a child who has just learned the checkerboard will reuse it. Hand out this page only after the map of Problem 21 is filled in.

Problem 24. (reserve)

Answer. 11 ways: 7 with a lying domino on the top-left square, 4 with a standing one.

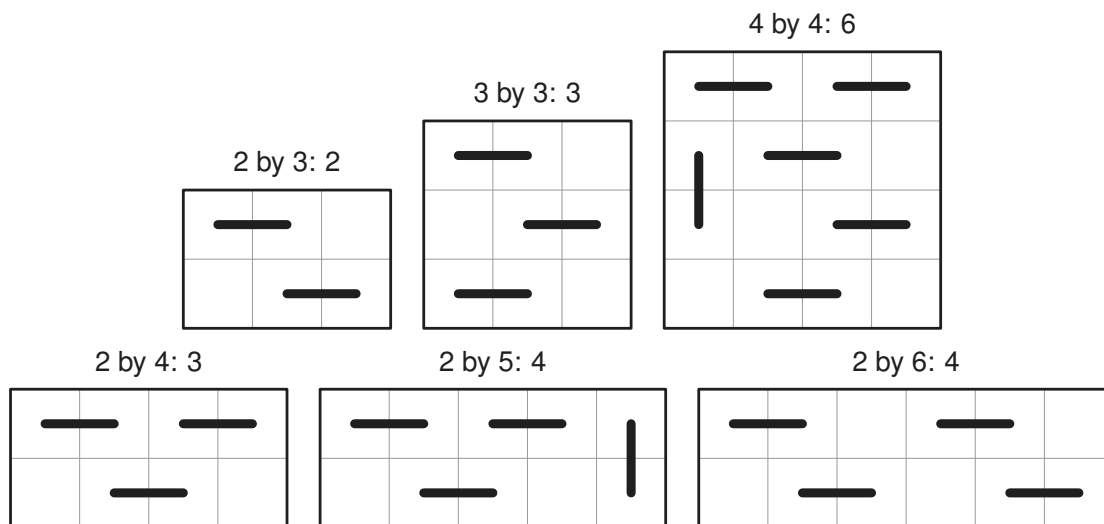




Argument. Sort by the top-left square, then within each group by what covers the square next to it; the pictures are in that order. A child who finds 9 or 10 should be sent back to the sorting: “which group is short?”

Problem 25–26. (reserve)

Answer. Fewest dominoes so that no more fit: 2 by 3: **2**; 3 by 3: **3**; 4 by 4: **6**; 2 by 4: **3**; 2 by 5: **4**; 2 by 6: **4**; 2 by 2: **2** (one domino leaves two neighbouring squares, so a second one always fits). So the row 2 by 2 to 2 by 6 reads 2, 2, 3, 4, 4. What to notice: the best 2 by 6 is the best 2 by 3 twice (a lying domino in the top row on columns 1–2 and one in the bottom row on columns 2–3 jam three columns with two dominoes, and the two free squares at the ends of the block are not neighbours of anything free in the next block), so 2 by 9 needs **6**. The values for 2 by 7, 8, 9, 10 are 5, 6, 6, 7 (computed): two dominoes per three columns, rounded up.



Argument. These minimum values were checked by computer. What a child can argue: on the 3 by 3, two dominoes leave 5 uncovered squares, no two of them side by side (else a domino would fit). The only five squares of a 3 by 3 with no two side by side are the corners and the centre (a light square such as the top middle rules out its three neighbours and leaves room for at most three more). So the four covered squares would be the four middle-of-side squares, no two of which are side by side, and no domino can lie on them. So 2 is impossible, and the picture does it with 3. On the 4 by 4, 4 dominoes leave 8 uncovered squares no two of which are neighbours, which forces them to be a whole colour class (Section 7, Problem 22); the 8 covered squares are then the other colour class, in which no domino can lie at all, so 4 is impossible. That 5 is also impossible has no short argument; it is a computer fact.

If the grades 2–3 table stalls. Problem 3 or 4: give the sorting by the left end (Problem 6) early. Problem 10: hand a child the tiling of board B and ask for board H. Problem 17: place the middle L for them. **If it finishes the core:** Problem 13, then 14–18, then boards E–H of Problem 10 and Problem 11, then 19–23, then 24–26. The share item for this table is Problem 8–9 (the map and the checkerboard reason).

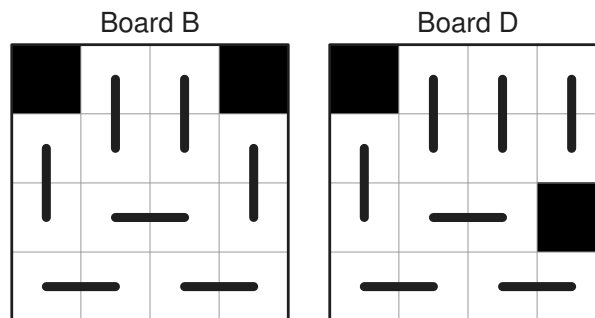
7. Grades 4–5 table

Order and timing (core, about 40 minutes). Problems 1 (6 min), 2 (3), 3 (4), 4 (3), 6 (1), 7 with pictures P and Q only (4), 8 (3), 11 (10; if the table is slow, do the 8 by 8 together at the share instead), 12 (4). This keeps the two share items, the chessboard of Problem 4 and the 8 by 8 of Problem 11, inside the hour. **Reserve, in this order:** 13–14 (13), 5 (6), 9–10 and 15 (8), 16 (8), 17–21 (25), 22–30 (about 60). Problem 11 (the 8 by 8 with 21 L's) takes ten minutes even with the recipe.

How to run it. Same rules as the other tables; this table writes its own reasons, and the adult should push every “no” to a sentence with numbers in it.

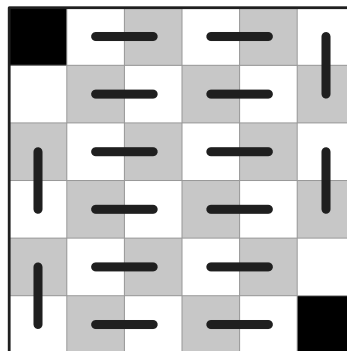
Problem 1.

Answer. A: **N**. B: **Y**. C: **N**. D: **Y**. (The 4–5 boards A, B, C, D are the 2–3 packet’s boards A, B, D, H.) Tilings of B and D:



Problem 2.

Answer. **N**. The most dominoes that can be placed is 16, leaving two light squares uncovered; children get 15 or 16.



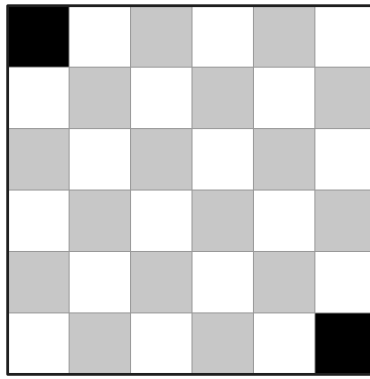
Problem 3.

Answer. Under a domino: always 1 dark and 1 light. Left to cover: A 6 dark, 8 light; B 7 and 7; C 6 and 8; D 7 and 7. The N boards are the ones whose two black squares have the same colour.

Argument. Seven dominoes cover 7 dark and 7 light squares, so a board with 6 dark and 8 light cannot be tiled. This is a complete reason for every N. (For the Y boards the tiling itself is the reason.)

Problem 4.

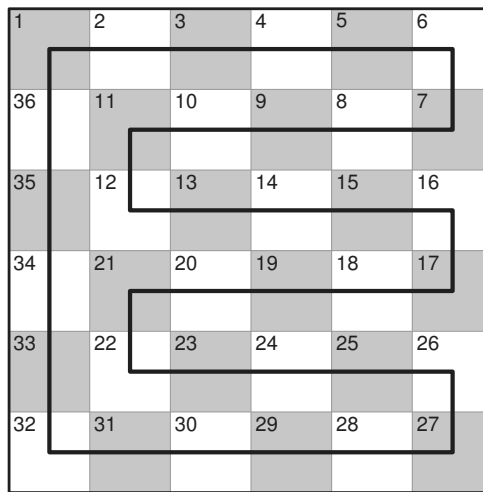
Answer. 6 by 6 minus opposite corners: 16 dark, 18 light, **N**. 8 by 8 minus opposite corners: 30 dark, 32 light, **N**: 31 dominoes would cover 31 dark and 31 light squares, but there are only 30 dark ones. Opposite corners always have the same colour.



Hints, in order. (1) “What colour are the two corners?” (2) “How many dark squares would 31 dominoes cover?”

Problem 5. (reserve)

Answer. With one dark and one light black square, both roads have an even number of squares, so dominoes laid along each road cover everything. Loop order, square 36 next to square 1 (this is the same loop as on the student page):

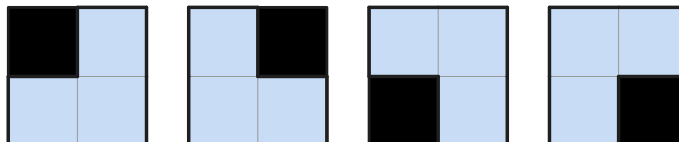


Argument. Colours alternate along the path. Between a dark square and a light square, going either way round, there is an even number of squares (the run starts and ends with the two colours in the right order: it starts with the opposite colour of the black square you left and ends with the opposite colour of the one you reach). Each even run is covered two squares at a time. So on this board *every* pair of one dark and one light square can be tiled around (all 324 such pairs, and the 306 same-colour pairs never), and the same path exists on the 8 by 8 and on every board with an even side. Together with Problem 3 this is the whole answer: two black squares, dominoes: possible if and only if the colours differ.

Hints, in order. (1) “Count the squares from one black square to the other along the path.” (2) “Why is the count even? What colour is the first square of the road, and the last?”

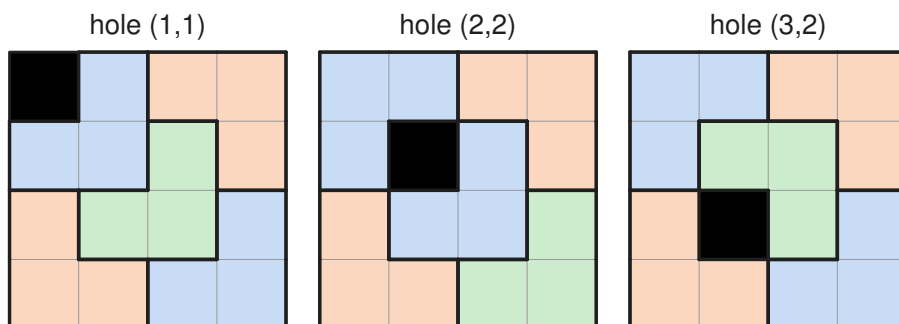
Problem 6.

Answer. One L each; the four places of the black square give the four positions of the L, which is the point: a 2 by 2 minus any square is one L.



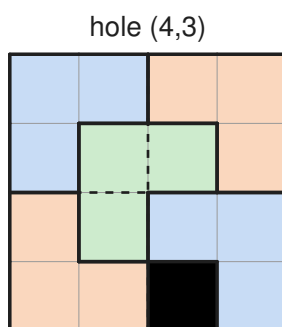
Problem 7. (P and Q core, R and S reserve)

Answer. Every square works, and for every black square there is exactly one tiling (computer check of all 16; the uniqueness is a fact to state, not to prove). Because each board has exactly one tiling, trial and error is genuinely slow: give the middle-L hint after three minutes. In the core do P and Q only; R and S are reserve. P (1,1), Q (2,2), R (3,2):



Problem 8.

Answer. T (black square at row 4, column 3) by the recipe:



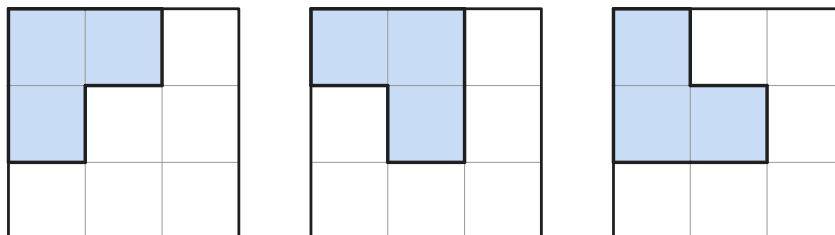
Argument. Cut the board into four 2 by 2 quarters; one holds the black square. One L in the very middle covers one square of each of the other three quarters. Now each quarter is a 2 by 2 minus one square: one L each. The recipe only needs to know which quarter the black square is in.

Problem 9. (reserve)

Answer. No.

Problem 10. (reserve)

Answer. Three positions; never two corners; four corners and three L's, so some corner is uncovered and Problem 9 is impossible.

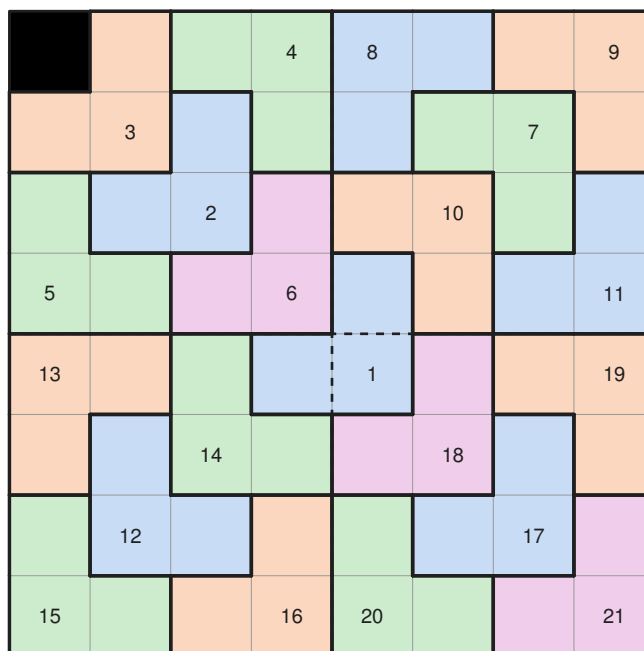


Argument. Every L lies inside a 2 by 2 block, and every 2 by 2 block of the 3 by 3 contains exactly one corner square.

Problem 11.

Answer. The 21 L's, numbered in the order the recipe places them (1 is the middle L; 2, 7, 12, 17 are the

middle L's of the four quarters):



Hints, in order. (1) “Which quarter has the black square? The middle L covers one square of each of the other three.” (2) “Now look at one quarter alone: it is a 4 by 4 with one square gone. Do Problem 8 inside it.”

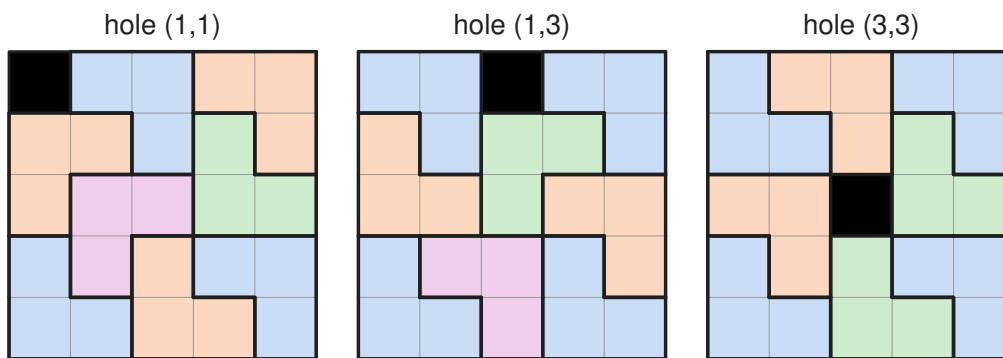
Problem 12.

Answer. The middle L covers one square of each of the three quarters without the black square; now each quarter is an 8 by 8 board with one square black or covered, and Problem 11 showed that the recipe handles any 8 by 8 with one square missing. So the 16 by 16 works, with $1 + 4 \times 21 = 85$ L's. The same sentences work for 32 by 32 (quarters are 16 by 16 boards with one square missing), with $1 + 4 \times 85 = 341$ L's ($32 \times 32 - 1 = 1023 = 3 \times 341$), and for 64 by 64, and so on.

Argument. This is induction: it works for 4 because it works for 2, for 8 because it works for 4, for 16 because it works for 8. A child's sentence: “each step is the same as the step before, one size up”.

Problem 13. (reserve)

Answer. Places are given as (row, column), counted from the top-left corner. Places 1 (1,1) Y, 2 (1,2) N, 3 (1,3) Y, 4 (2,2) N, 5 (2,3) N, 6 (3,3) Y. The map: Y exactly on the 9 squares whose row and column are both odd; N on the other 16. (Numbers of tilings: corner 8, middle of a side 16, centre 32.)

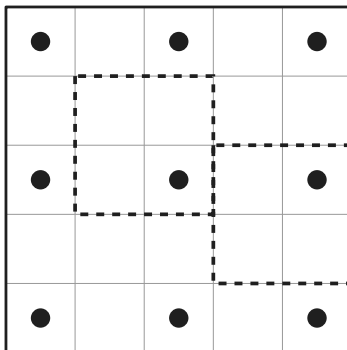


Notes. Children guess from the 3 by 3 and 4 by 4 that “everything works” or “corners and centre”; places 3 and 4 kill both. Square (2,2) is dark on the checkerboard but fails for L's: the checkerboard is not the

right colouring for L's.

Problem 14. (reserve)

Answer. Every 2 by 2 square contains exactly 1 dot; an L covers at most 1 dot; eight L's cover at most 8 dots; there are 9; so the black square must be on a dot. This is exactly the map of Problem 13.



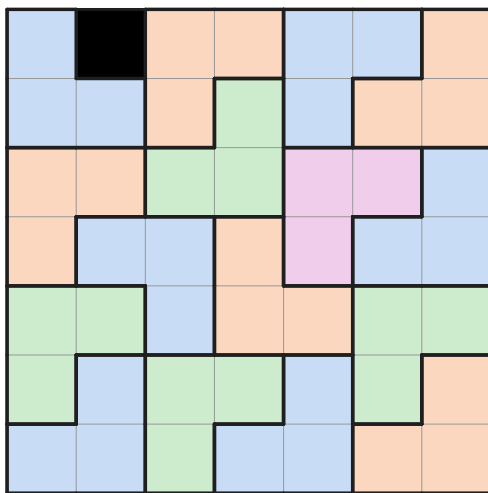
Argument. Of two neighbouring rows exactly one is odd, and of two neighbouring columns exactly one is odd, so each 2 by 2 window holds exactly one (odd, odd) square. Every L fits inside a 2 by 2 window. If the black square were not a dot, the 8 L's would have to cover all 9 dots.

Problem 15. (reserve)

Answer. At most 1 dot per L; at most 3 for three L's; 4 dots; impossible. The 3 by 3 dots are its corners, so this is Problem 10 again.

Problem 16. (reserve)

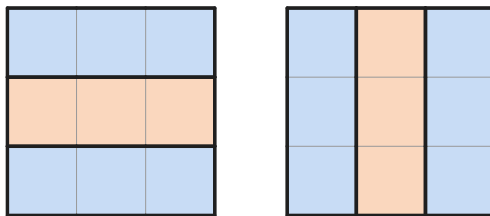
Answer. 16 dots, 16 L's: the dot argument says nothing (16 L's could cover 16 dots, one each), so it does not say N. And indeed **Y**: a tiling exists.



Notes. The rule from 5 by 5 ("only odd, odd") does not continue: on the 7 by 7 every one of the 49 squares works as the black square (checked by computer; state it, do not ask the table to verify it). The dot argument only bites when there are more dots than pieces, which happens for 5 by 5 (9 dots, 8 pieces) and 3 by 3 (4 dots, 3 pieces) but not for 7 by 7 (16 and 16). General fact for the mathematicians: an $n \times n$ board minus one square, $3 \nmid n$, is L-tileable for every hole except when $n = 5$ (Chu–Johnsonbaugh 1986).

Problem 17. (reserve)

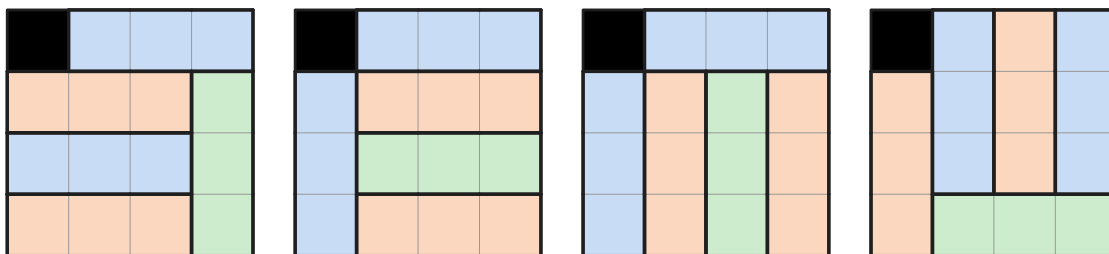
Answer. 2 ways: three lying bars, or three standing bars.



Argument. If the top-left square is under a lying bar, that bar is the whole top row; the square below it cannot start a standing bar (that would need three rows below the top), so row 2 is a lying bar, and then row 3. If it is under a standing bar, the same by columns. Two tilings and no others.

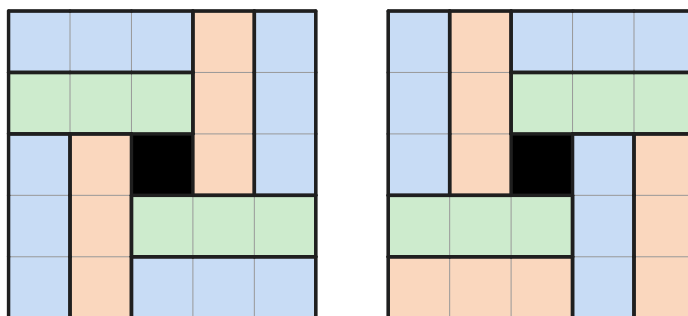
Problem 18. (reserve)

Answer. Place 1 (corner) **Y**, place 2 (1,2) **N**, place 3 (2,2) **N**. The map: Y only at the four corners (4 tilings each; the four tilings for the top-left corner are shown), N on the other 12.



Problem 19. (reserve)

Answer. Places 1–5 **N**, place 6 (the centre) **Y**. The map: Y only at the centre (2 tilings, the pinwheel and its mirror), N on the other 24.



Problem 20. (reserve)

Answer. A straight tromino, lying or standing, covers the numbers 1, 2, 3 (one of each) on either numbering. First numbering: six 1s, five 2s, five 3s. Five trominoes cover five of each number, so the black square must carry a 1. Second numbering: also six 1s, so again a 1. Squares that are 1 in both numberings: exactly the four corners. That matches the map of Problem 18 completely: the corners work (Problem 18's tilings) and nothing else can.

first				second			
1	2	3	1	1	3	2	1
2	3	1	2	2	1	3	2
3	1	2	3	3	2	1	3
1	2	3	1	1	3	2	1

Argument. On the first numbering, three squares in a row or in a column carry three consecutive numbers on the diagonals, hence 1, 2, 3 in some order; the same for the second numbering. Counting the numbers on the 4 by 4 gives 6, 5, 5. One numbering alone only shows that the black square is one of six squares; the second numbering cuts the six down to the four corners.

Hints, in order. (1) “Put a tromino down; read the numbers under it. Turn it. Move it.” (2) “How many of each number do five trominoes cover?” (the page has a blank for this: 5, 5, 5; against 6, 5, 5 on the board, the odd one out is the black square’s number). (3) “Which squares are 1 in the first picture? Which of those are 1 in the second?”

Problem 21. (reserve)

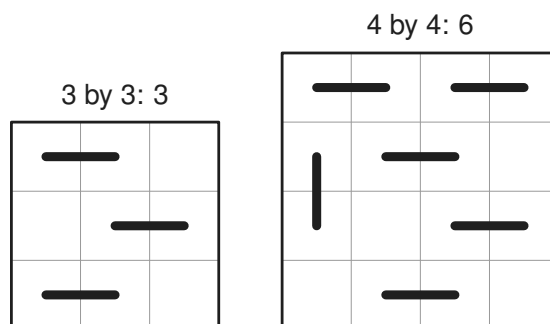
Answer. Each numbering of the 5 by 5 has eight 1s, nine 2s, eight 3s. Eight trominoes cover eight of each, so the black square must carry a 2 in the first numbering and a 2 in the second. The only square that is a 2 in both is the centre. This matches the map of Problem 19: the centre works (Problem 19’s tilings) and nothing else can.

first					second				
1	2	3	1	2	2	1	3	2	1
2	3	1	2	3	3	2	1	3	2
3	1	2	3	1	1	3	2	1	3
1	2	3	1	2	2	1	3	2	1
2	3	1	2	3	3	2	1	3	2

Notes. Same-size piece as the L, opposite answer: the straight tromino is the strict one (4 by 4: corners only; 5 by 5: centre only; 8 by 8: only the four squares (3,3), (3,6), (6,3), (6,6), Golomb 1954), while the L is generous (every square on 4 by 4, 7 by 7 and 8 by 8). The two colourings are the tool.

Problem 22. (reserve)

Answer. 3 by 3: 3 dominoes. 4 by 4: 6. The page asks only why 4 fails on the 4 by 4 (the uncovered squares must be one whole colour class of the checkerboard, and then no domino can lie on the other class); if a child asks about 5, say that 5 was ruled out by computer and no short reason is known.



Argument. 3 by 3: two dominoes leave 5 uncovered squares, no two of them neighbours (or a domino would still fit). The only 5 squares of a 3 by 3 with no two neighbours are the corners and the centre: a middle-of-side square rules out its three neighbours and leaves room for at most three more, so an uncovered set of five contains no middle-of-side square. Then the 4 covered squares are the four middle-of-side squares, no two of which are neighbours, and no domino can lie on them. So 2 is impossible and the picture does 3. 4 by 4: 4 dominoes leave 8 uncovered squares, no two neighbours; each row then

holds exactly 2 of them, in columns $\{1, 3\}$, $\{1, 4\}$ or $\{2, 4\}$ (two uncovered squares in a row of four with no two side by side); the pattern $\{1, 4\}$ clashes with every pattern in the row below (a square directly above an uncovered one), so consecutive rows alternate $\{1, 3\}$ and $\{2, 4\}$, which is a whole colour class of the checkerboard; the 8 covered squares are then the other colour class, in which no two squares are neighbours, so no domino can lie there. Hence 4 is impossible. That 5 is also impossible is a computer check; there is no short argument, and it is honest to say so.

Problem 23. (reserve)

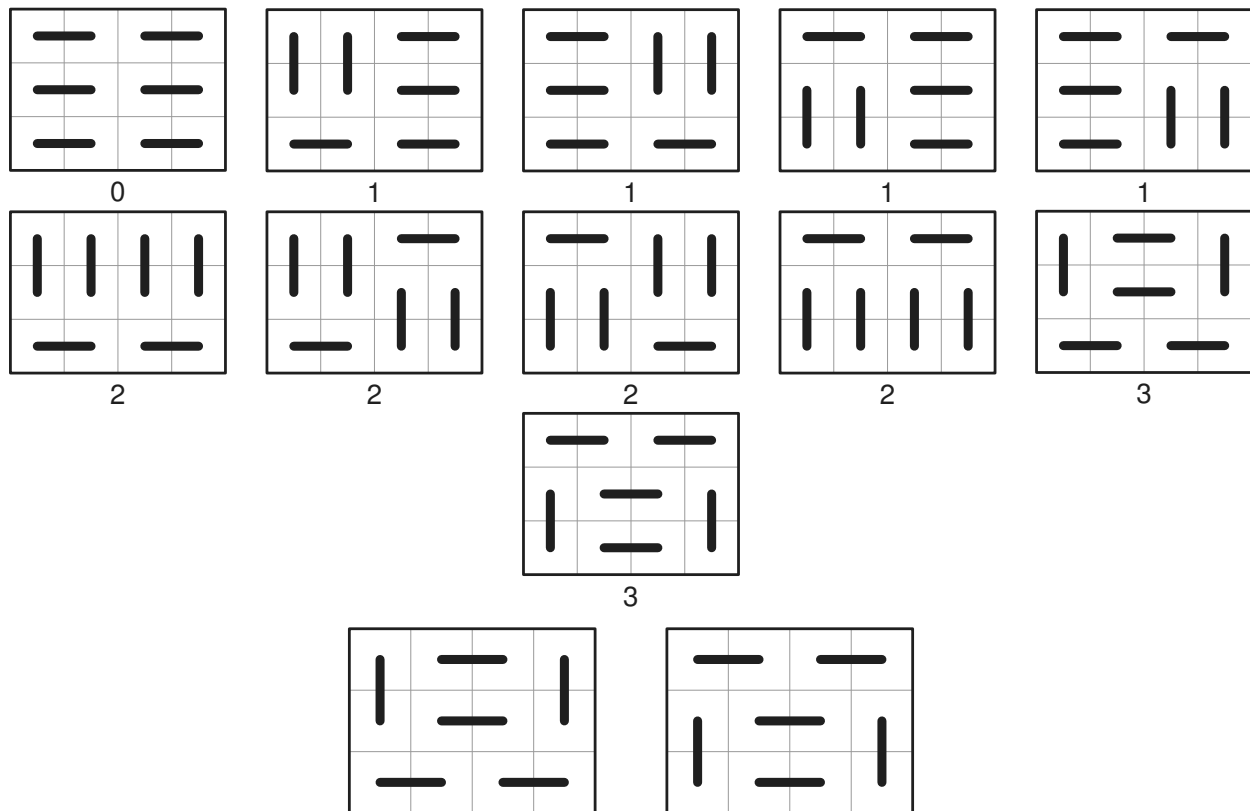
Answer. 2, 3, 5, 8; then 13, 21 (each is the sum of the two before). On the 2 by 5: 5 ways with a standing domino at the left end and 3 with two lying dominoes, $5 + 3 = 8$: the two counts are the two previous numbers. The pictures are in Section 6, Problems 1–4, with the argument under Problem 6 there.

Problem 24. (reserve)

Answer. 11: 7 with a lying domino at top left, 4 with a standing one. Pictures in Section 6, Problem 24.

Problem 25. (reserve)

Answer. From the all-lying tiling, the fewest flips to each of the 11 tilings are 0, 1, 1, 1, 1, 2, 2, 2, 2, 3, 3 (the largest is 3). The two tilings farthest apart are 6 flips apart:



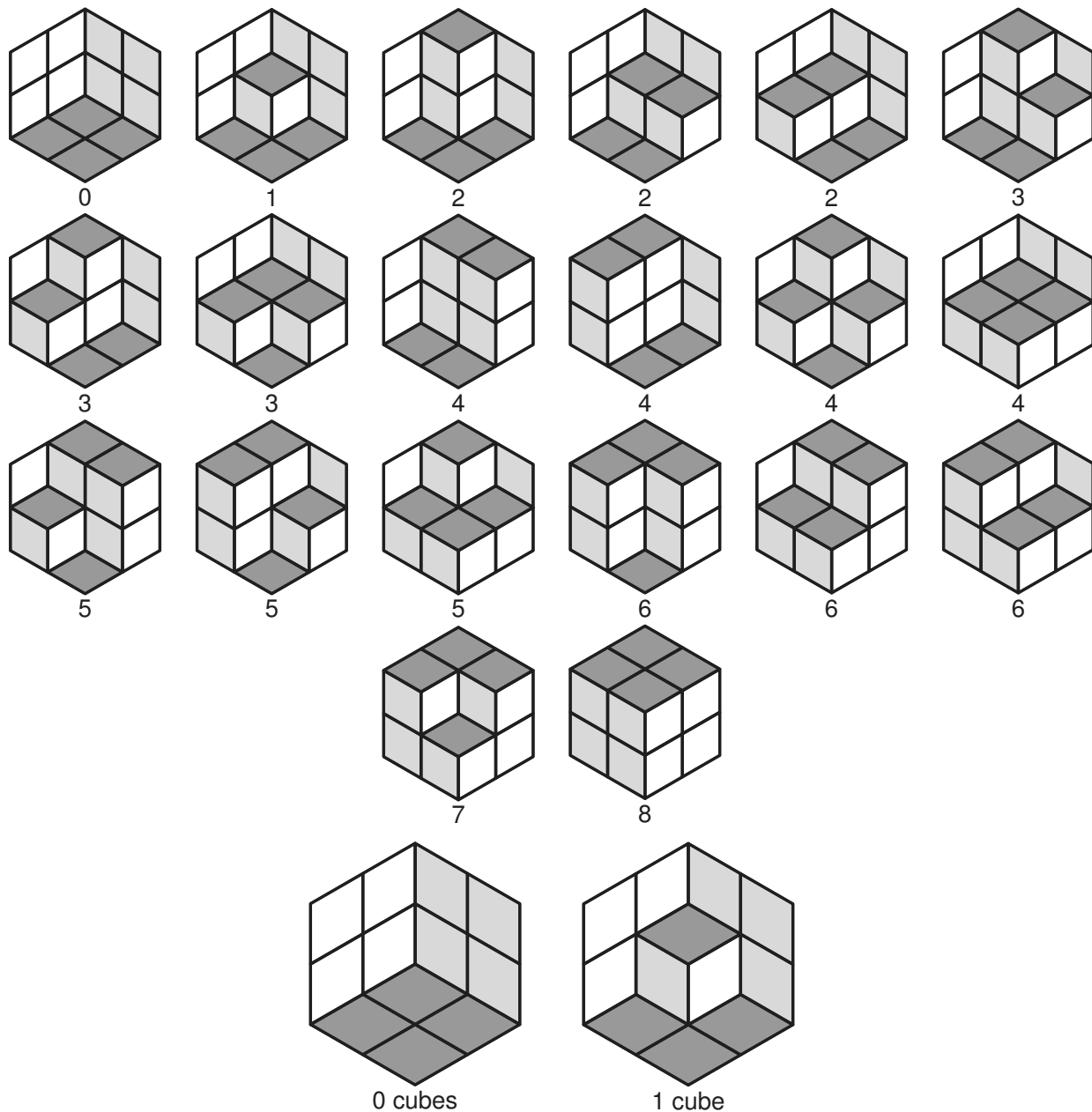
Notes. Every tiling can be reached from every other by flips (true for every rectangle, and for every region without holes: Thurston 1990). The 36 tilings of the 4 by 4 are connected too, with farthest distance 10.

Problem 26. (reserve)

Answer. 12 rhombi; always **4 standing, 4 leaning left, 4 leaning right**; 20 fillings in all. The argument (chains from the bottom edge) is under K–1 Problem 13 in Section 5.

Problem 27. (reserve)

Answer. All 20 fillings, turned a quarter turn as the page says, with the standing rhombi shaded as cube tops and the number of cubes under each; a spin of three rhombi around a point adds or removes exactly one cube, and the fewest spins from the empty corner to the full 2 by 2 by 2 box is **8**.



Notes. The same picture flips in the eye between a pile in a corner and a hollow, so the page fixes the reading: after the quarter turn, all shaded rhombi at the bottom is the empty box (0 cubes, the first picture above) and all shaded rhombi at the top is the full box (8, the last picture); with that reading the counts are the ones under the pictures, and every spin changes the count by one. The 20 fillings are the ways to stack cubes in a 2 by 2 by 2 corner so that the stacks do not get taller going away from the corner (plane partitions); the spin graph is connected, and the two extremes are exactly 8 spins apart.

Problem 28. (reserve)

Answer. Straight tetrominoes: **N**, although $36 = 9 \times 4$. L-tetrominoes: **N**.

Problem 29. (reserve)

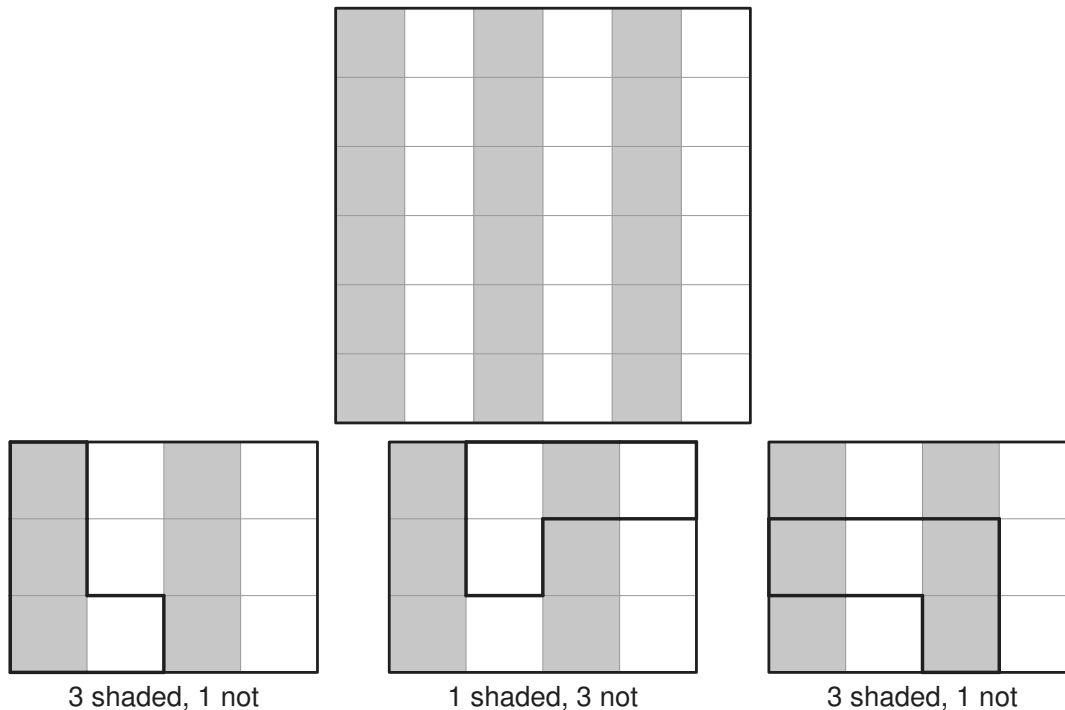
Answer. A straight tetromino, lying or standing, covers 1, 2, 3, 4 (one of each). The board has nine 1s,

ten 2s, nine 3s and eight 4s; nine bars would cover nine of each, but there are ten 2s and eight 4s; so N.

1	2	3	4	1	2
2	3	4	1	2	3
3	4	1	2	3	4
4	1	2	3	4	1
1	2	3	4	1	2
2	3	4	1	2	3

Problem 30. (reserve)

Answer. An L-tetromino covers 3 shaded squares and 1 unshaded, or 1 shaded and 3 unshaded, always an odd number of shaded squares (three in a column and the foot in the next column; or three in a row, two of one colour and one of the other, and the foot under an end square of the row, the same colour as that end); nine odd numbers add to an odd number; the board has 18 shaded squares, even; so N.



If the grades 4–5 table stalls. Problem 2: stop at three minutes and go to Problem 3, the reason is the point. Problem 11: place the middle L and the first quarter's middle L for them. Problems 20–21: do one placement of a tromino with them and ask them to read the numbers aloud. **If it finishes the core:** 13–14, 5, 9–10 and 15, 16, 17–21, 22–30 in that order; the sheet (c) pieces are only needed for 28. The share item for this table is Problem 4 (the chessboard) and the 8 by 8 of Problem 11.

8. The share (last eight minutes)

Each table gets two minutes; the organizer asks for one finding and one reason, and the children say the reason.

- K–1: the pinwheel (three reds fill the side-3 triangle in two ways) and “blues cannot fill the big triangle, even though 16 is even: there are 10 mountains and 6 valleys and a blue always eats one of each”. If Problem 13 was done: “however you fill the big hexagon with blues, it is always 4, 4, 4”.
- Grades 2–3: the 3 by 3 map (corners and centre work, sides do not) and the checkerboard reason: “a domino eats one dark and one light; there are 5 dark and 4 light; the black square must be dark”.
- Grades 4–5: the chessboard with two opposite corners removed (“30 dark, 32 light, 31 dominoes: no”) and the 8 by 8 with 21 L’s by the recipe, with the sentence “it works for 8 because it works for 4”.
- Organizer, one sentence to close: three tables, three kinds of pieces, one idea: count two kinds of cells; every piece eats a fixed amount of each; if the numbers do not match, no tiling can exist, and no amount of trying will change that. And when the numbers do match, you still have to find the tiling, which is a different job.

9. When a table stalls or finishes early

- **Stalled on a “yes”** (cannot find the tiling): give the first piece, placed correctly, and ask for the second. Every yes-tiling is drawn above.
- **Stalled on a “no”** (still trying): stop the trying after three starts and move to the recording and the colouring page. The colouring or numbering page is the argument; the child’s job is to place a piece on it and read what is under it.
- **Stalled on “every way”**: ask what covers the top-left square (or the bottom edge, for pattern blocks) and sort the found ways by the answer; the missing ones show up as a short group.
- **Two children disagree** (one says can, one says cannot): the one who says can shows it; if they cannot show it, the table looks for the reason together.
- **Finished the core**: K–1 goes to Problems 13, 15, 17; grades 2–3 to 13, 14–18, then boards E–H of 10 and 11, then 19–23, 24–26; grades 4–5 to 13–14, 5, 9–10 and 15, 16, 17–21, 22–30. A child who wants to keep building can cut more pieces and try the last board of the packet at any time.
- **Finished everything**: for grades 4–5, the 6 by 6 with two black squares of the child’s choice and the loop; then “which squares of the 8 by 8 can be black for straight trominoes” (answer: only (3,3), (3,6), (6,3), (6,6), counted as (row, column) from the top-left corner. Number both diagonals as on the student pages, so that the top-left square carries 1 in the first numbering and 2 in the second; then each numbering has 21, 22, 21 squares of the numbers 1, 2, 3, so the black square must carry a 2 in both numberings, and exactly these four squares do; all four work). For grades 2–3: the 4 by 4 with L’s and one black square anywhere, drawn without the recipe, then with it. For K–1: the side-2 hexagon with reds only (8 reds, 9 fillings).

10. The mathematics behind the session

The questions (for the two mathematicians). Every problem in the packets is an instance of one of these: existence (can this region be tiled by these pieces?), impossibility by an invariant, completeness (is the invariant also sufficient?), counting, reconfiguration, optimality, and listing shapes. A tiling found is a proof of yes; a failed attempt proves nothing, which is why the invariants exist.

Invariants. Weight the cells so that every placed piece consumes a fixed amount; if the region's total is wrong, no tiling exists. The session uses four costumes of this one idea: the checkerboard (dominoes; dark minus light must be 0), up- and down-pointing triangles (blue rhombi; a triangle of side n has n more up-triangles than down, so no triangle outline is ever rhombus-tileable, even side 4 with 16 cells), the dots on (odd, odd) squares (L-trominoes; an L covers at most one dot), and the two diagonal 3-colourings (straight trominoes; a bar covers one of each colour). Beyond colourings: Conway–Lagarias (1990) tiling groups and boundary-word invariants, which prove for instance that the triangular array T_N of hexagonal cells is tileable by tribones iff $N \equiv 0, 2, 9, 11 \pmod{12}$, a fact no colouring gives; Pak's ribbon-tile invariants (2000); Thurston's height functions (1990).

Completeness. Gomory's loop (Section 6, Problem 13; Honsberger, *Mathematical Gems I*, 1973): remove one dark and one light square from an even board and a domino tiling always exists. Odd boards minus one square: tileable iff the hole is dark (snake path). Deficient $2^n \times 2^n$ boards with L-trominoes: always (Golomb 1954, the induction of Problems 8–12). Chu–Johnsonbaugh (1986): every deficient $n \times n$ board with $3 \nmid n$ is L-tileable except $n = 5$, where exactly the (odd, odd) holes work; and an $m \times n$ rectangle ($m, n \geq 2$) is L-tileable iff $3 \mid mn$ and it is not 3 by odd. Straight trominoes on the deficient 8 by 8: exactly the four squares (3,3), (3,6), (6,3), (6,6) (Golomb). Domino tilings are perfect matchings of the grid graph, so existence is decidable in polynomial time; for L-trominoes the existence question is NP-complete (Moore–Robson 2001), for bars too (Beauquier–Nivat–Rémila–Robson 1995), and for Wang tiles on the whole plane it is undecidable (Berger 1966), which is where aperiodic tilings and the 2023 aperiodic monotile of Smith–Myers–Kaplan–Goodman–Strauss come from.

Counting. $2 \times n$: Fibonacci, by the left-end argument. $3 \times n$: 3, 11, 41, 153 for $n = 2, 4, 6, 8$, with $a_n = 4a_{n-2} - a_{n-4}$. Kasteleyn and Temperley–Fisher (1961) count every rectangle by a determinant: 4×4 has 36 domino tilings, 6×6 has 6,728, 8×8 has 12,988,816. MacMahon's box formula counts rhombus tilings of hexagons of side 1, 2, 3: 2, 20, 980; they are plane partitions in an $n \times n \times n$ box, and the spin graph is a distributive lattice (Problem 27). Aztec diamonds have $2^{n(n+1)/2}$ tilings (Elkies–Kuperberg–Larsen–Propp 1992) and random tilings have limit shapes (the arctic circle, Jockusch–Propp–Shor 1998).

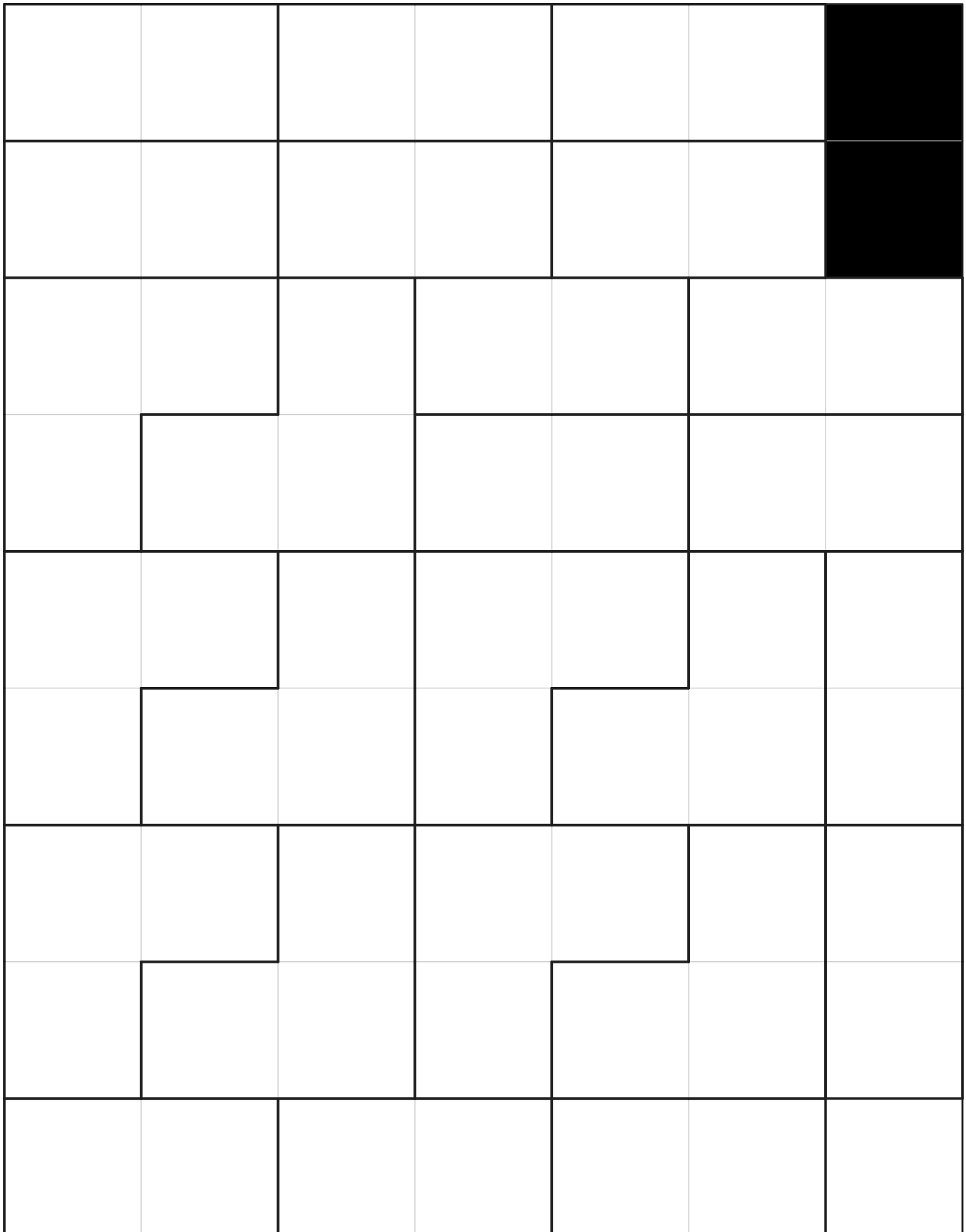
Reconfiguration. Any two domino (or rhombus) tilings of a simply connected region are joined by local flips, and the height function measures the distance (Thurston 1990). The 11 tilings of the 3 by 4 have flip diameter 6; the 20 rhombus tilings of the side-2 hexagon have spin diameter 8, attained only by the empty and full boxes.

Optimality. “Fewest dominoes so that no more fit” is a minimum maximal matching (edge domination), NP-hard even for planar bipartite graphs of maximum degree 3 (Yannakakis–Gavril 1980); the small values used here (2, 2, 3, 4, 4 for $2 \times n$; 3 for 3×3 ; 6 for 4×4) are computed, not from a formula; for $2 \times n$ the values up to $n = 10$ (2, 2, 3, 4, 4, 5, 6, 6, 7) equal $\lceil 2n/3 \rceil$, verified only that far.

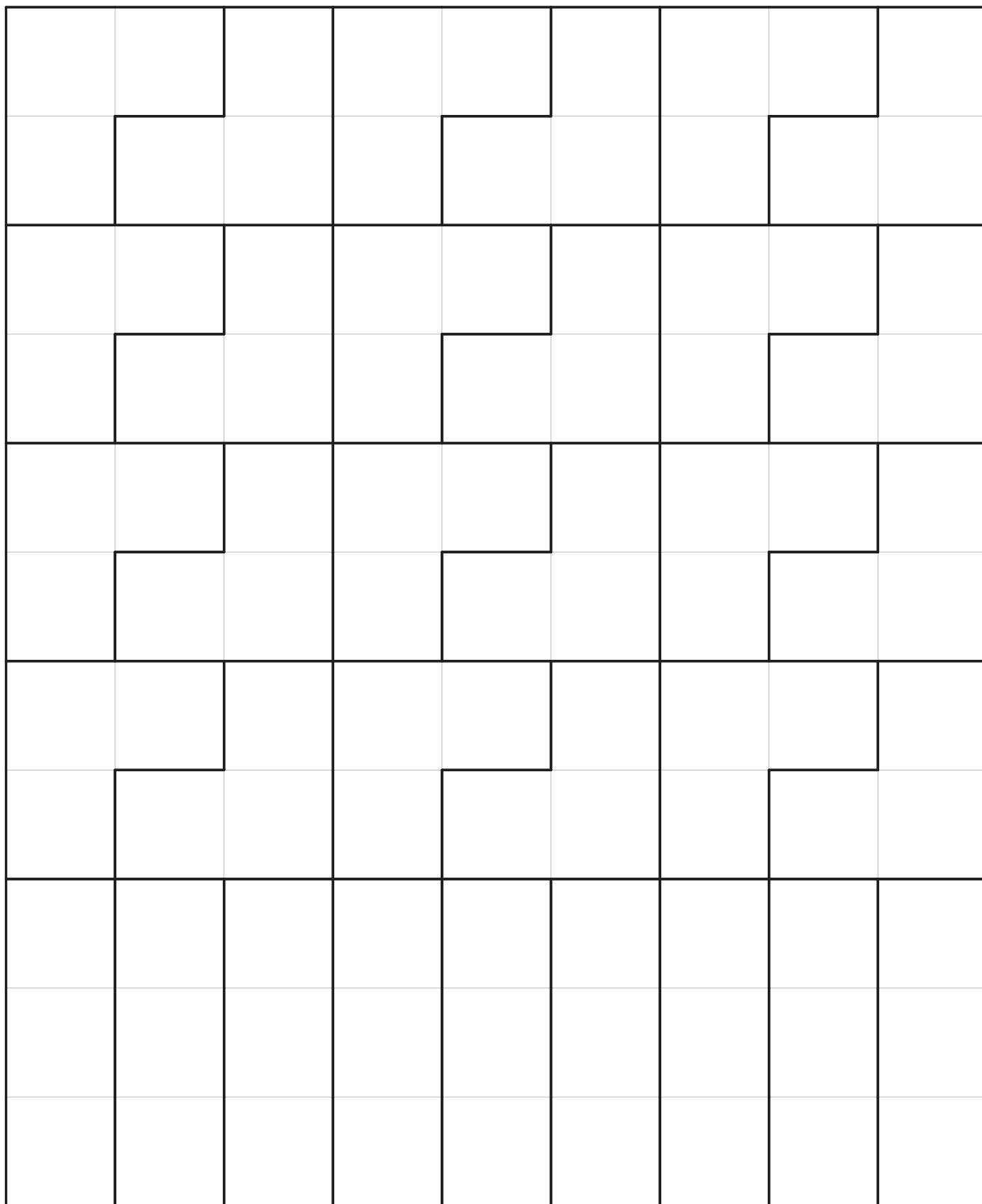
Shapes. Free polyiamonds: 1, 1, 1, 3, 4, 12, 24, 66, ... (OEIS A000577), no closed form; the up/down count is the rhombus case of the height function.

Surveys. Ardila–Stanley, “Tilings”, Math. Intelligencer 32 (2010); Pak, “Tile invariants: new horizons”, Theoret. Comput. Sci. 303 (2003); Golomb, *Polyominoes* (1994); Kenyon's lecture notes on the dimer model.

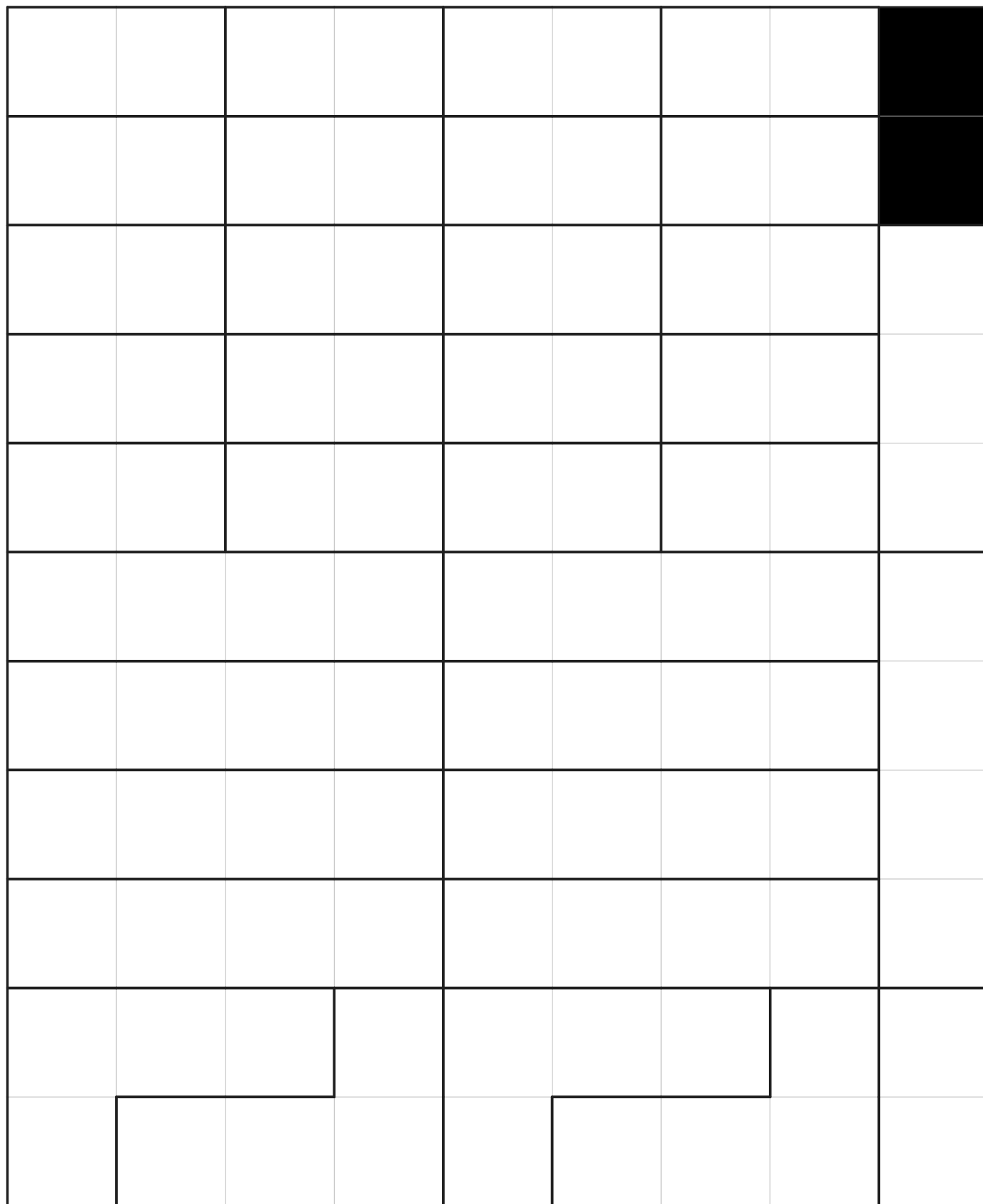
Cut sheet, 1 inch (grades 2–3 and K–1). Print at 100%, one per child (five copies). Cut along the thick lines: 15 dominoes, 10 L's, 2 black squares; the last square is waste.



Cut sheet (a), 0.8 inch (grades 4–5). Print at 100%, one per child (three copies). Cut along the thick lines: 24 L's and 9 straight trominoes.



Cut sheet (b), 0.8 inch (grades 4–5). Print at 100%, one per child (three copies). Cut along the thick lines: 21 dominoes, 2 black squares, 1 straight tromino, 9 straight tetrominoes, 4 L-tetrominoes.



Cut sheet (c), 0.8 inch (grades 4–5, needed for Problem 28 only). Cut along the thick lines: 12 L-tetrominoes, 3 dominoes, 6 L's, 9 straight trominoes.

